RAJALAKSHMI ENGINEERING COLLEGE

Unit I – Introduction to Data Science

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1. EVOLUTION OF DATA SCIENCE

❖ What is called data?

- Data Latin word → datum (singular) →which means the "thing given".
- Data is any information that has been translated into different forms to be processed, analyzed, managed, and transferred.
- Data refers to **raw information that consists of basic facts and figures**.
- In computer era data definition would then include any raw information that is stored on a computer, such as numerical data, images, coding, notes, and financial data.



Fig 1.1: Comparison of Data and Information

There are two types of data: Qualitative and Quantitative data, which are further classified into

four categories:

1. Nominal data
2. Ordinal data
3. Discrete data
4. Continuous data

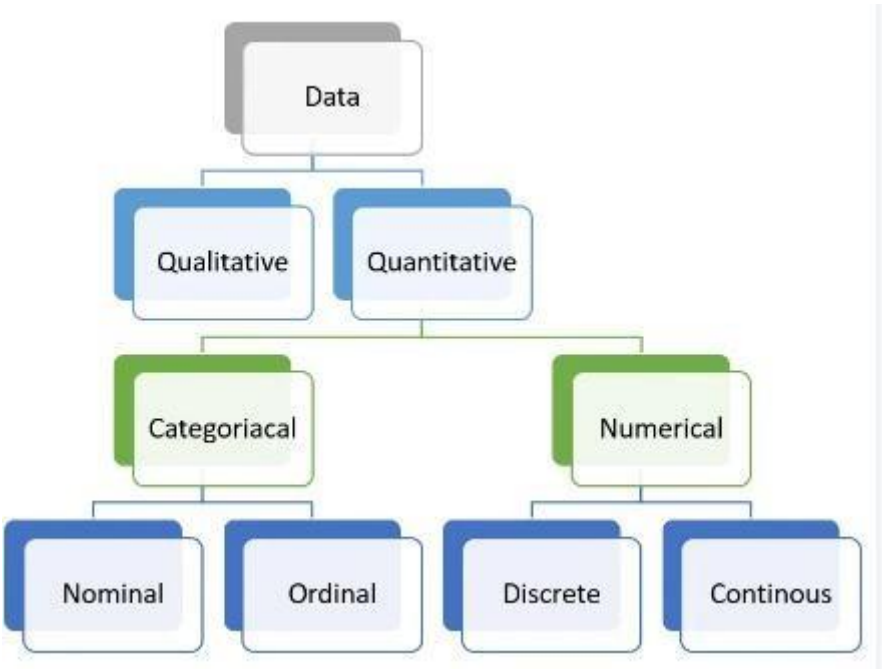


Fig 1.2: Data Classification

1. Qualitative or Categorical Data

Qualitative or Categorical Data is a type of data that **can't be measured or counted in the form of numbers.**

These types of **data are sorted by category, not by number**. That's why it is also known as Categorical Data. These data consist of audio, images, symbols, or text. The gender of a person, i.e., male, female, or others, is qualitative data.

Qualitative data tells about the perception of people. This data helps market researchers understand the customers' tastes and then design their ideas and strategies accordingly.

The other examples of qualitative data are :

- What language do you speak
- Favorite holiday destination
- Opinion on something (agree, disagree, or neutral)
- Colors

The Qualitative data are further classified into two parts :

A. Nominal Data

Nominal Data is used to label variables without any order or quantitative value. The color of hair can be considered nominal data, as one color can't be compared with another color.

The name "nominal" comes from the Latin name "nomen," which means "name." With the help of nominal data, we can't do any numerical tasks or can't give any order to sort the data. These data don't have any meaningful order; their values are distributed into distinct categories.

Examples of Nominal Data :

- Colour of hair (Blonde, red, Brown, Black, etc.)
- Marital status (Single, Widowed, Married)
- Nationality (Indian, German, American)
- Gender (Male, Female, Others)
- Eye Color (Black, Brown, etc.)

B. Ordinal Data

Ordinal data have natural ordering where a **number is present in some kind of order by their position on the scale**. These data are used for observation like customer satisfaction, happiness, etc., but we can't do any arithmetical tasks on them.

Ordinal data is qualitative data for which their values have some kind of relative position. These kinds of data can be considered "in-between" qualitative and quantitative data.

The ordinal data only shows the sequences and cannot use for statistical analysis. Compared to nominal data, ordinal data have some kind of order that is not present in nominal data.

Examples of Ordinal Data :

- When companies ask for feedback, experience, or satisfaction on a scale of 1 to 10
- Letter grades in the exam (A, B, C, D, etc.)
- Ranking of people in a competition (First, Second, Third, etc.)
- Economic Status (High, Medium, and Low)
- Education Level (Higher, Secondary, Primary)

❖ Difference between Nominal and Ordinal Data

Table 1.1: Difference between Nominal and Ordinal Data

Nominal Data	Ordinal Data
Nominal data can't be quantified, neither they have any intrinsic ordering	Ordinal data gives some kind of sequential order by their position on the scale
Nominal data is qualitative data or categorical data	Ordinal data is said to be "in-between" qualitative data and quantitative data
They don't provide any quantitative value, neither can we perform any arithmetical operation	They provide sequence and can assign numbers to ordinal data but cannot perform the arithmetical operation

Nominal data cannot be used to compare with one another	Ordinal data can help to compare one item with another by ranking or ordering
Examples: Eye color, housing style, gender, hair color, religion, marital status, ethnicity, etc	**Examples:** Economic status, customer satisfaction, education level, letter grades, etc

2. Quantitative Data

Quantitative data is a type of data that can be **expressed in numerical values**, making it countable and including statistical data analysis. These kinds of data are also known as Numerical data.

It answers the questions like "how much," "how many," and "how often." For example, the price of a phone, the computer's ram, the height or weight of a person, etc., falls under quantitative data.

Quantitative data can be used for statistical manipulation. These data can be represented on a wide variety of graphs and charts, such as bar graphs, histograms, scatter plots, boxplots, pie charts, line graphs, etc.

Examples of Quantitative Data :

- Height or weight of a person or object
- Room Temperature
- Scores and Marks (Ex: 59, 80, 60, etc.)
- Time

The Quantitative data are further classified into two parts :

A. Discrete Data

The term discrete means distinct or separate. The discrete data contain the values that fall under **integers or whole numbers.** The total number of students in a class is an example of discrete data. These data can't be broken into decimal or fraction values.

The discrete data are **countable and have finite values; their subdivision is not possible**. These data are represented mainly by a bar graph, number line, or frequency table.

Examples of Discrete Data :

- Total numbers of students present in a class
- Cost of a cell phone
- Numbers of employees in a company
- The total number of players who participated in a competition
- Days in a week

B. Continuous Data

Continuous data are in the form of **fractional numbers**. It can be the version of an android phone, the height of a person, the length of an object, etc. Continuous data represents information that **can be divided into smaller levels**. The continuous variable can take any value within a range.

The key difference between discrete and continuous data is that **discrete data contains the integer** or **whole number**. Still, **continuous data stores the fractional numbers** to record different types of data such as temperature, height, width, time, speed, etc.

Examples of Continuous Data :

- Height of a person
- Speed of a vehicle
- "Time-taken" to finish the work
- Wi-Fi Frequency
- Market share price

❖ Difference between Discrete and Continuous Data

Table 1.2: Difference between Discrete and Continuous Data

Discrete Data	Continuous Data
Discrete data are countable and finite; they are whole numbers or integers	Continuous data are measurable; they are in the form of fractions or decimal
Discrete data are represented mainly by bar graphs	Continuous data are represented in the form of a histogram
The values cannot be divided into subdivisions into smaller pieces	The values can be divided into subdivisions into smaller pieces
Discrete data have spaces between the values	Continuous data are in the form of a continuous sequence
Examples: Total students in a class, number of days in a week, size of a shoe, etc	**Example:** Temperature of room, the weight of a person, length of an object, etc

❖ Data Science:

- ✓ Data Science is a **combination of mathematics, statistics, machine learning, and computer science.** Data Science is **collecting, analyzing and interpreting data to gather insights** into the data that can help decision-makers make informed decisions.

- ✓ Data Science is used in almost every industry today that can predict customer behavior and trends and identify new opportunities. Businesses can use it to make informed decisions about product development and marketing. It is used as a tool to detect fraud and optimize processes. Governments also use Data Science to improve efficiency in the delivery of public services.

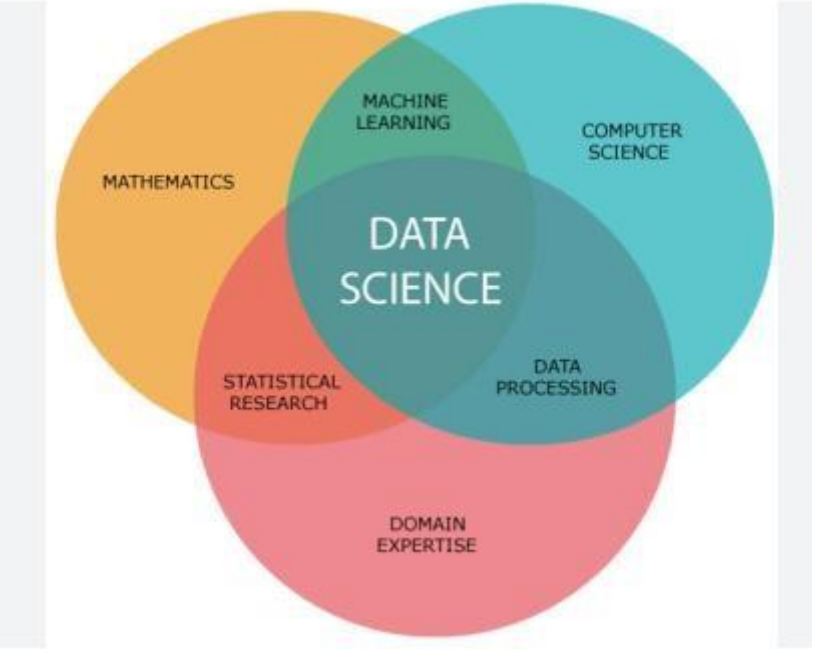


Fig 1.3: Data Science Representation

- ✓ Data Science is an emerging field that sees its importance grow with each passing day. It is the latest buzzword in the IT world, and its demand in the market has been growing steadily. The demand for Data Scientists is proliferating, driven by the need for **organizations to transform data into insights**. Companies such as **Google, Amazon, Microsoft, and Apple** have been among the biggest recruiters of Data Scientists. Data Science is also becoming a sought-after field for IT professionals.

- ✓ According to a recent report by Precedence Research, the demand for Data Science is anticipated to grow at a CAGR (Compound Annual Growth Rate) of 16.43% and hit the market value of a whopping 378.7 Billion over the forecast period 2022 to 2030.

- ✓ In simple terms, Data Science **helps to analyze data and extract meaningful insights** from it by combining statistics & mathematics, programming skills, and subject expertise.

➢ Importance of Data Science

Nowadays, organizations are overwhelmed with data. Data Science will help in extracting meaningful insights from that by combining various methods, technology, and tools. In the fields of e-commerce, finance, medicine, human resources, etc, businesses come across huge amounts of data. Data Science tools and technologies help them process all of them.

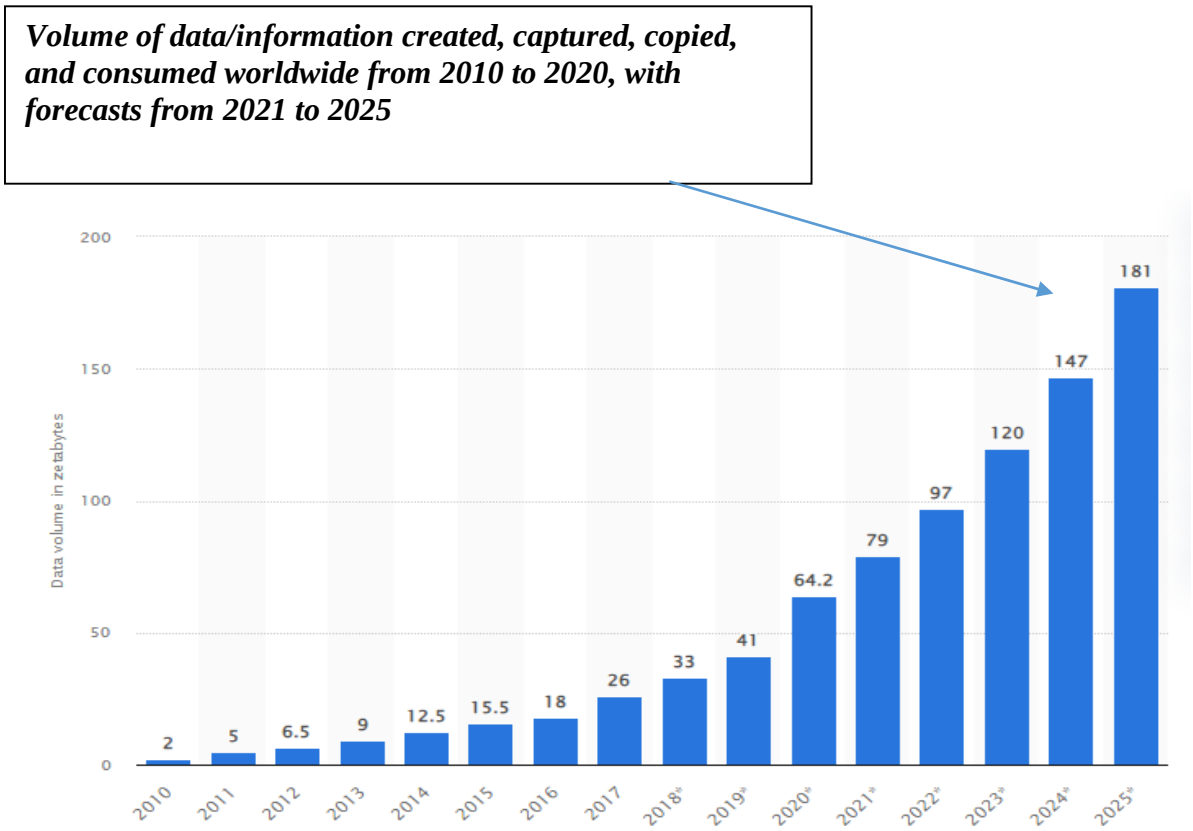


Fig 1.4: Data Creation Statistics

➢ History of Data Science

Early in the 1960s, the term "Data Science" was coined to help comprehend and analyze the massive volumes of data being gathered at the time. Data science is a discipline that is constantly developing, employing computer science and statistical methods to acquire insights and generate valuable predictions in a variety of industries.

➢ Data Science – Prerequisites

➢ Statistics

Data science relies on statistics to capture and transform data patterns into usable evidence through the use of complex machine-learning techniques.

Check out Statistics for Data Science to learn key concepts of Statistics in Data Science, Machine Learning, and Business Intelligence.

- **Programming**

Python, R, and SQL are the most common programming languages. To successfully execute a data science project, it is important to instill some level of programming knowledge.

Check out the free R Studio Tutorial course to learn how to use various techniques and tools to process the raw data and obtain valuable insights.

- **Machine Learning**

Making accurate forecasts and estimates is made possible by Machine Learning, which is a crucial component of data science. You must have a firm understanding of machine learning if you want to succeed in the field of data science.

- **Databases**

A clear understanding of the functioning of Databases, and skills to manage and extract data is a must in this domain.

- **Modeling**

You may quickly calculate and predict using mathematical models based on the data you already know. Modeling helps in determining which algorithm is best suited to handle a certain issue and how to train these models.

❖ Data Science Evolution:

Data science has evolved significantly over the past few decades, transforming from basic data analysis into a sophisticated, multidisciplinary field that encompasses ***statistics, machine learning, data mining, and big data technologies.*** Here is a detailed overview of the evolution of data science, illustrated with examples:

1. Early Days of Data Analysis (1960s-1970s)

- **Statistics and Mathematics:** The foundations of data science lie in statistics and mathematics. In the 1960s and 1970s, data analysis primarily involved statistical methods for hypothesis testing and data collection.
- A hypothesis is an **assumption that is made based on some evidence**. This is the initial point of any investigation that **translates the research questions into predictions**. It includes components like variables, population and the relation between the variables.
- **Example:** The use of statistical quality control methods in manufacturing, pioneered by W. Edwards Deming , to improve product quality and production efficiency.

The Deming Wheel

Often called the PDSA Cycle or the Shewhart Cycle, the Deming Cycle is an exceptionally simple model for improvement. Deming learned this technique from his colleague Dr. Walter Shewhart and made an effort to popularize the method.

There are four parts to this method, they are as follows:



Fig 1.5: Deming Wheel

- **Plan –** Those implementing the change must have a plan for how to complete the change as well as how to collect any necessary data. What is the objective? What will happen and why? How will the change be tested?
- **Do** – Try the new method out on a smaller scale. What were the problems? Analyze any data gathered.
- **Study** – Study the results! How does it compare to the predictions made in the Plan step? How would you summarize the test? What have you learned?
- **Act** – Based on what was learned from the test, now is the time for refinement. What modifications should be made? Loop back around to prepare planning for the next test if needed.

2. Database Management and Business Intelligence (1980s-1990s)

- **Relational Databases:** The development of relational database management systems (RDBMS) by companies like IBM (with their DB2 product) allowed for more efficient storage, retrieval, and management of data. A relational database (RDB) is a **way of structuring information in tables, rows, and columns.** An RDB has the ability to establish links—or relationships–between information by joining tables, which makes it easy to understand and gain insights about the relationship between various data points. A few examples of RDBMSs include MySQL, Oracle Database, and SQL Server.

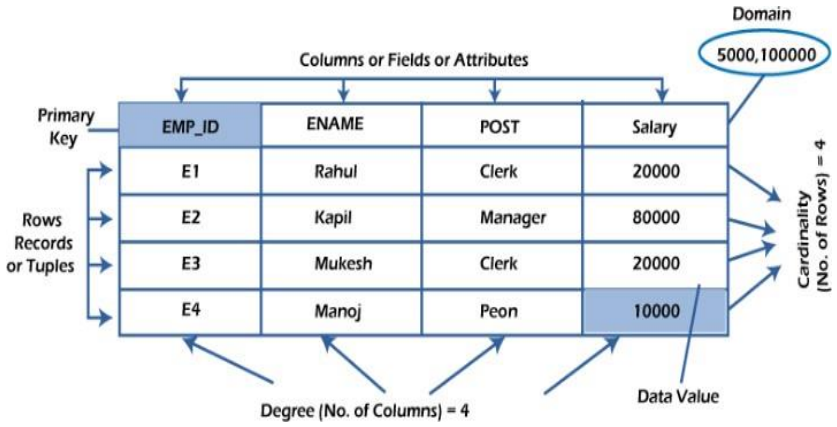


Fig 1.6: Relational DataBase Example

- **Business Intelligence (BI):** Tools like SAS (**Statistical Analysis System**) and IBM's DB2 enabled businesses to perform complex queries and generate reports, leading to the rise of BI.



. **Fig 1.7: IBM DB2 Icon**

- SAS is an analytics software used by a number of sectors, including healthcare, finance and retail. It is used for advanced analytics, data management and business intelligence.

- **Example:** Walmart's implementation of data warehousing and BI tools to analyze sales data and optimize inventory management.

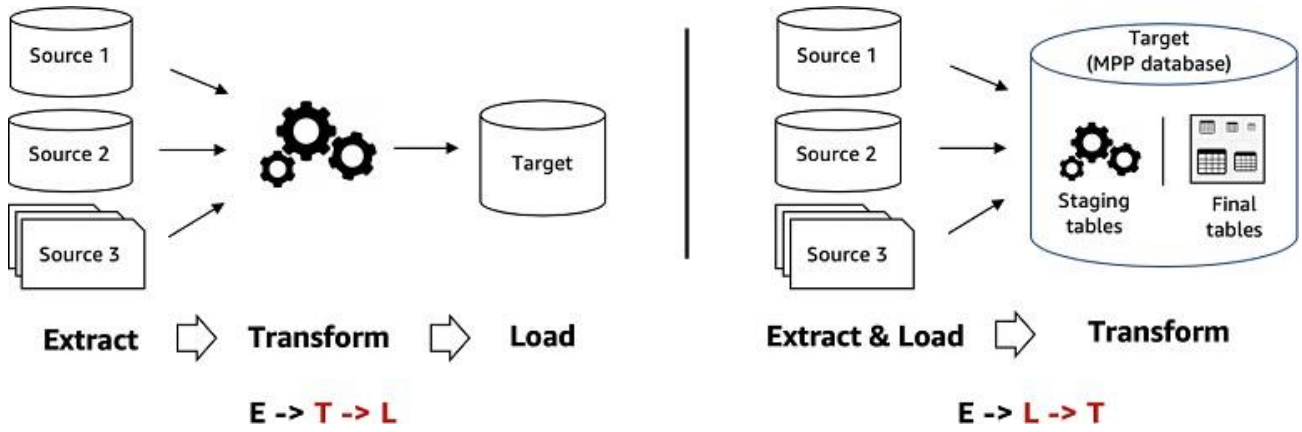
3. Rise of Data Warehousing and Online Analytical Processing (OLAP) (1990s)

- **Data Warehousing:** Technologies like data warehousing emerged, allowing for the storage of large volumes of historical data for analysis. ETL (Extract, Transform, Load) processes became standard for integrating data from various sources.



. **Fig 1.8: Data ware House**

- **What is ETL and ELT?**
  - ✓ Extract, transform, and load (ETL) and extract, load, and transform (ELT) are two data-processing approaches for analytics.
  - ✓ Large organizations have several hundred (or even thousands) of data sources from all aspects of their operations—like applications, sensors, IT infrastructure, and third-party partners. They have to filter, sort, and clean this large data volume to make it useful for analytics and business intelligence.
  - ✓ The ETL approach uses a set of business rules to process data from several sources before centralized integration.



. **Fig 1.9: ETL Vs ELT**

- ✓ The ELT approach loads data as it is and transforms it at a later stage, depending on the use case and analytics requirements.
- ✓ The ETL process requires more definition at the beginning. Analytics must be involved from the start to define target data types, structures, and relationships.
- ✓ Data scientists mainly use ETL to load legacy databases in the data warehouse, while ELT has become the norm today.

Table 1.3: Comparison between ETL and ELT

Category	ETL	ELT
s	Extract, transform, and load	Extract, load, and transform
Process	Takes raw data, transforms it into a predetermined format, then loads it into the target data warehouse.	Takes raw data, loads it into the target data warehouse, then transforms it just before analytics.
Transformation and load locations	Transformation occurs in a secondary processing server.	Transformation takes place in the target data warehouse.
Data compatibility	Best with structured data.	Can handle structured, unstructured, and semi-structured data.
Speed	ETL is slower than ELT.	ELT is faster than ETL as it can use the internal resources of the data warehouse.
Costs	Can be time-consuming and costly to set up depending on ETL tools used.	More cost-efficient depending on the ELT infrastructure used.
Security	May require building custom applications to meet data protection requirements.	You can use built-in features of the target database to manage data protection.

- **OLAP:** Online Analytical Processing allowed for multidimensional analysis of data, enabling more dynamic and flexible data exploration.
- Online Analytical Processing (OLAP) is a method for creating queries from multidimensional data, primarily for delivering insights for Business Intelligence.
- OLAP involves three core operations: aggregation / consolidation (roll-up), drill-down (from summary to detail), and slicing and dicing (snapshots across multiple dimensions).
- OLAP contrasts with the other method for data processing, OLTP or Online Transaction Processing, which focuses on processing individual transactions in real time. For example, OLTP processes (captures, stores, updates) sales transactions in real time whereas OLAP analyzes (aggregates, distills, investigates) the trend in sales transactions over time.
- **(i) Relational OLAP (ROLAP)** – Queries made to data stored in relational databases (data stored in individual tables that can be combined in a query)
- **(ii) Multidimensional OLAP (MOLAP)** – Queries made to data store in a multidimensional cube (data already combined and stored with combinations readily available for queries)
- **(iii) Hybrid OLAP (HOLAP)** – Queries made to a data store that combines relational with multidimensional.
-
- **Example: Amazon's use of data warehousing** and **OLAP** to analyze customer purchasing behavior and recommend products.

Table 1.4: Comparison of Data warehouse,Data Lake and Data Marts

DATA WAREHOUSES VS DATA LAKES VS DATA MARTS

	Data warehouses	Data lakes	Data marts
Usage	The data analysis and reporting needs of an entire organization	The reporting needs of different kinds and difficulty, predictive analytics	The reporting needs of a specific operational department or subject
Data stored (typically)	Larger volumes of structured data; processed	Huge volumes of structured and unstructured data; raw	A limited amount of structured data; processed
Data sources	An array of external and internal sources, covering different areas of business	Any external or internal sources	Few sources linked to one business area
Size	Larger than 100 GB	Larger than 100 GB	Smaller than 100 GB
Ease of creation	Difficult to set up	Difficult to set up	Easy to set up

4. Advent of Big Data (2000s)

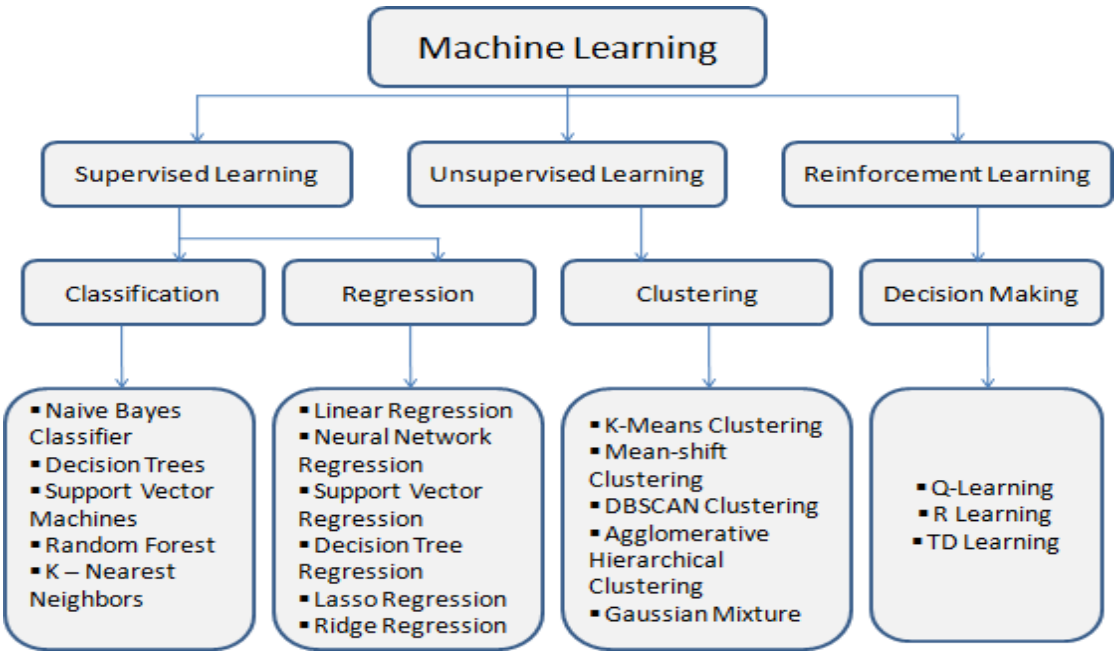
- **Big Data Technologies:** With the explosion of data from the internet, social media, and IoT devices, traditional databases couldn't handle the volume, velocity, and variety of data. Technologies like Hadoop and NoSQL databases (e.g., MongoDB, Cassandra) were developed.
- **SQL Vs NoSQL:**

✓ SQL→Query Language for storing, manipulating and retrieving data stored in table format.Ex: MySQL,Oracle,Sqlite,PostgreSQL and MS-SQL etc.
✓ NoSQL→ Provides mechanism for storage and retrieval of data other than tabular relations model used in relational databases (Non relational databases).Ex: MongoDB, BigTable,RavenDB,Cassandra,Hbase,Neo4j , CouchDB etc.

- **Example:** Facebook's use of Hadoop to process petabytes of data generated by user interactions and to deliver personalized content.

5. Emergence of Machine Learning and AI (2010s)

- **Machine Learning:** Advances in machine learning algorithms and the availability of large datasets enabled more sophisticated predictive models and AI applications. Tools like TensorFlow and PyTorch became popular.
- **What is Machine Learning?** Machine learning (ML) is a branch of artificial intelligence (AI) and computer science that focuses on the using data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.
- **Example:** Google's use of machine learning for search engine algorithms, speech recognition, and image classification.



. **Fig 1.10: Machine Learning Categories**

Supervised learning:

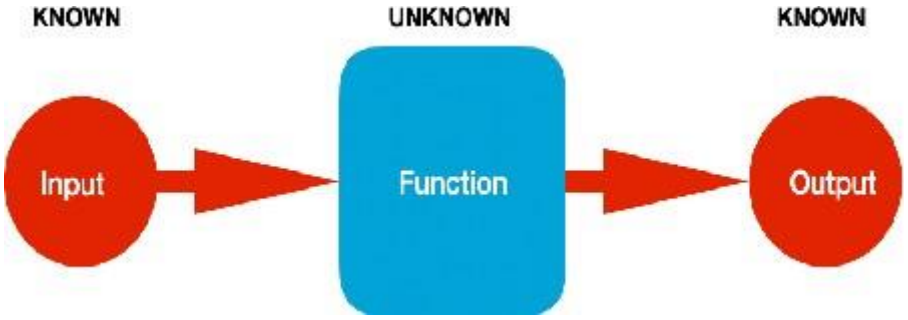


Fig 1.11: Supervised Learning Model

Supervised Learning is a category in which we feed labelled data as input to the machine learning model. Supervised learning, also known as supervised machine learning, is a subcategory of machine learning and artificial intelligence. It is defined by its use of labeled data sets to train algorithms that to classify data or predict outcomes accurately.

As input data is fed into the model, it adjusts its weights until the model has been fitted appropriately, which occurs as part of the cross validation process. Supervised learning helps organizations solve for a variety of real-world problems at scale, such as classifying spam in a separate folder from your inbox. It can be used to build highly accurate machine learning models.

Supervised learning uses a training set to teach models to yield the desired output. This training dataset includes inputs and correct outputs, which allow the model to learn over time. The algorithm measures its accuracy through the loss function, adjusting until the error has been sufficiently minimized.

From Fig 1.11 input and output values are already known, and the machine learning algorithm learns the mapping function from input to output. Mathematically, for Y as the Output and X as the input, machine learning algorithms try to find the best mapping function f such that Y = f(X). If you observe closely, Learning happens like some supervisor supervises the learning process. We already know the answers, so the algorithm tries to map the function so that the predicted output must be close to the actual output.

Let's say the machine has learned a mapping function f predicting values Y' for every X passed as an input to the function. Once the difference between predicted (Y') and actual (Y) goes below a certain threshold (in simple terms, errors become negligible),learning stops.

Supervised Learning can be further categorized as follows:

1. Classification: Taking the example of the following image Fig 1.12, images of shapes are the input to the machine learning model, and labels of those images as shape names are the output data. The labeling process is called "annotation". Based on these input and output data, the model learns to predictthe category of unseen image data, whether a rectangle, circle, triangle, or hexagon.

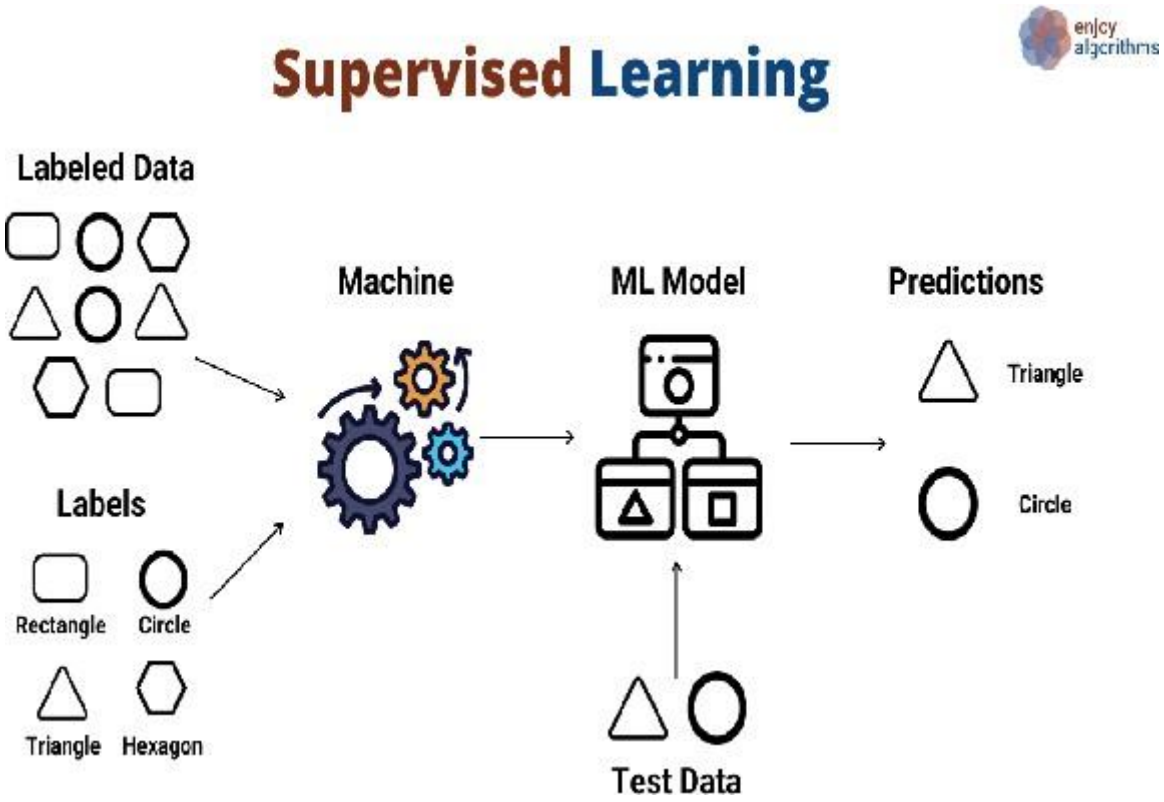


Fig 1.12: Supervised Learning – Classification Model

2. Regression: Example of the following image Fig 1.13 shows experience (in years) on the X-axis. For every experience, there is one salary (per month Rupees) on the Y-axis. Greendots are the Input and Output data coordinates (X, Y). So regression problem tries tofind the continuous mapping function from input to output variables, for example, theblue line in the image. If the order of the mapping function is fixed to 1, which is a linearfunction, the model will learn the blue line shown in the image.

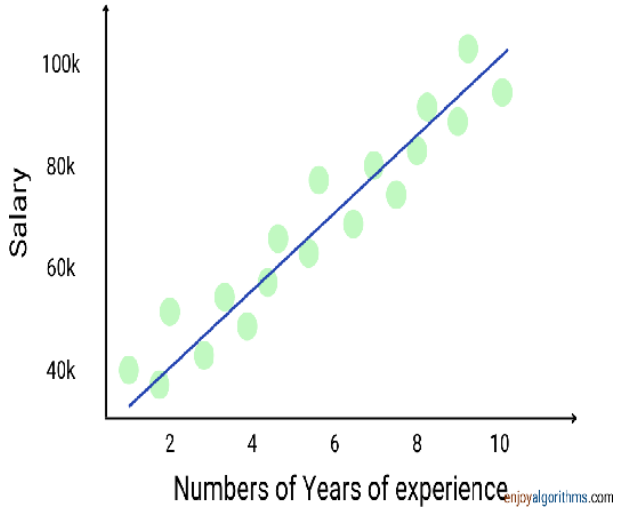


Fig 1.13: Supervised Learning – Classification Model

Some famous use cases of Supervised Learning are:

Object detection and Image classification: Find whether a cat is present in theimage or not. If yes, then find the location of the cat in that image.
Recommendation systems: If someone has bought a new home, automaticallysuggest new furniture as most people bought them together.
Time series prediction: If the last three days' atmosphere temperature of India was21°C, 22°C, and 21°C, then what would be the temperature for tomorrow?

Some frequently used algorithms in Supervised Learning

- Linear Regression and Logistic Regression
- SVM (Support Vector Machines)
- Random Forest

Unsupervised Learning

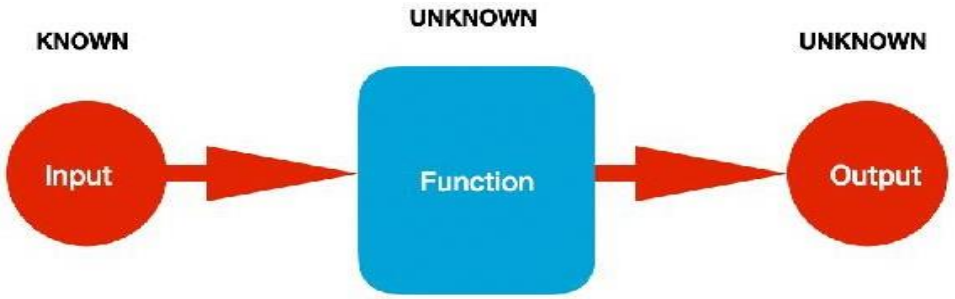


Fig 1.13: Unsupervised Learning Model

- **Unsupervised Learning** is a category of machine learning in which we only have the inputdata to feed to the model but no corresponding output data.
- Unsupervised learning, also known as unsupervised machine learning, uses machine learning (ML) algorithms to **analyze and cluster unlabeled data sets**. These algorithms **discover hidden patterns or data groupings without the need for human intervention.**
- Unsupervised learning's ability to discover similarities and differences in information make it the ideal solution for exploratory data analysis, cross-selling strategies, customer segmentation and image recognition.
- Unsupervised learning models are utilized for three main tasks—**clustering, association, and dimensionality reduction.**
- Here, we **know the value of input data**, but the output and mapping functions are both **unknown.** In such scenarios, machine learning algorithms map the function that finds similarity among different input data instances (samples) and groups them based on the similarity index, the output for unsupervised Learning. We can say that algorithms generate a pseudo output for learning the mapping function.
- In such Learning, there **is no supervision as there is no existence of output data. Hence they are called Unsupervised Learning**. Algorithms try to find similarity between different input

data instances by themselves using a defined similarity index. One of the similarity indexes can be the distance between two data samples to sense whether they are close or far.

- **Unsupervised Learning can further be categorized as:**

1. Clustering (Unsupervised classification): Using the example below, we have input data consisting of images of different shapes. Machine learning algorithms try to find the similarity among other images based on the colour pixel values, size, and shapes and form the groups as outputs in which similar input instances lie. If you notice, squares get clustered together, and similarly, circles and hexagons.

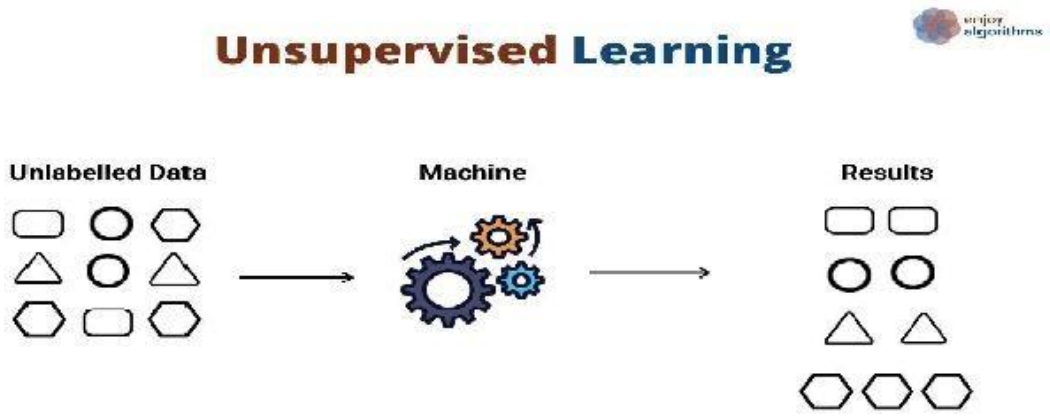


Fig 1.14: Unsupervised Learning-Clustering Model

Clustering Algorithms are:

- Hierarchical clustering
- K-means clustering

2. Dimensionality Reduction: When attributes of the data samples have more than three dimensions, there is no way to visualize the relationship among attributes, as wecan not plot variables in more than 3 Dimensions. But without analyzing the input data,we can never be sure about the machine learning model's performance. To solve this problem, we use dimensionality reduction techniques to reduce the total dimensions and analyze the data. Suppose we want to study 10 features together but cannot visualize the 10D plots. So, we try to reduce the dimensions to 3 or lower to easily plot it, analyze their relationship and do further processing.

Dimensionality Reduction Algorithms are:

- PCA (Principal Component Analysis)
- t-SNE

3. Association: Such Learning is more about discovering rules that describe a large portion of the data. Customers who bought a banana also bought carrots, and customers who bought a new house also bought new furniture. At first, it will look similar to clustering, but clustering is about finding the relationship among data points, and association is about finding the relationship among attributes/features of those data points.

Example Association Algorithm is:

Apriori Algorithm

Apriori algorithm is given by R. Agrawal and R. Srikant in 1994 for finding frequent itemsets in a dataset for boolean association rule. Name of the algorithm is Apriori because it uses prior knowledge of frequent itemset properties.

Some famous use cases of Unsupervised Learning are:
Market segmentation: Whether the market is hot or cold is based on the moneyrevolving in the market.
Fraud detection: Categorize transactions into fraudulent and non-fraudulent groups.
Image segmentation: Gray colour objects in an image taken from a car representsroads.

Some frequently used algorithms in Unsupervised Learning:

- K-means
- Apriori Algorithm for learning association rule.
- Principal Component Analysis

Semi-Supervised Learning

- Semi-supervised Learning is a category of machine learning in which we have input data, and only some input data are labelled. In more technical terms, the data is partially annotated.

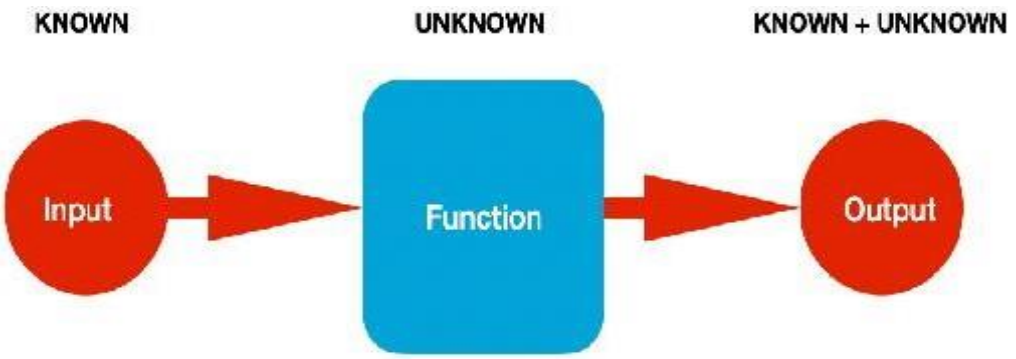


Fig 1.15: Semisupervised Learning Model

- Semi-supervised Learning is partially supervised and partially unsupervised. Let's take one example from the below image to make it clear. Suppose a bucket consists of three fruits, apple, banana, and orange. Someone captured the image of all three but labelled only the orange and banana images. Here, the machine will first classify the new apple image as not a banana or orange. Then someone will observe these predictions and label them as apples. Then retraining the model with that label will allow it to classify apple images as an apple.

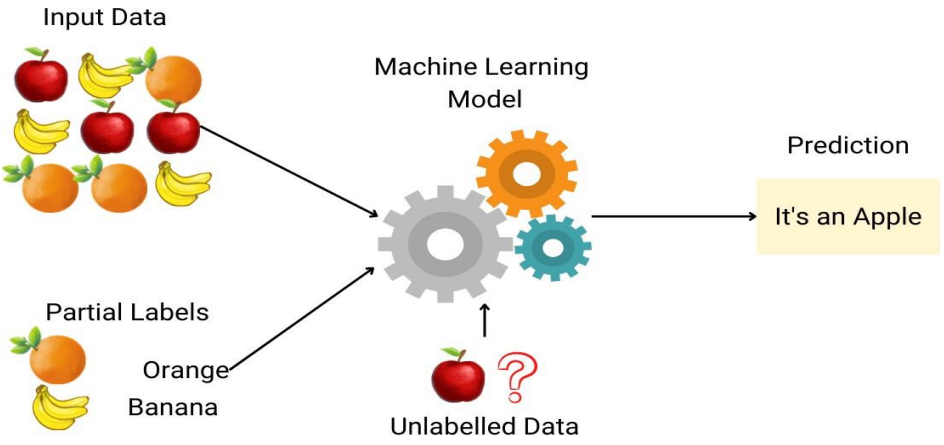


Fig 1.16: Semisupervised Learning-Example

- Nowadays, capturing a tremendous amount of data has become a trend. Many bigcompanies have collected millions of Terabytes of data and are still collecting. Butlabelling collected data requires a workforce and resources, so it's too expensive. This isthe main reason that many real-life databases fall into this category.

Some famous use cases of Semi-supervised Learning are:

In such types of Learning, one can use either supervised Learning or unsupervisedlearning methods.

- Supervised Learning: Train a model on labelled data and then use that Learning topredict output for unlabeled data. This output for the corresponding unlabeled input datacan now be treated as the pseudo label. Now we can use partial and pseudo-labelleddata to train another supervised learning model.

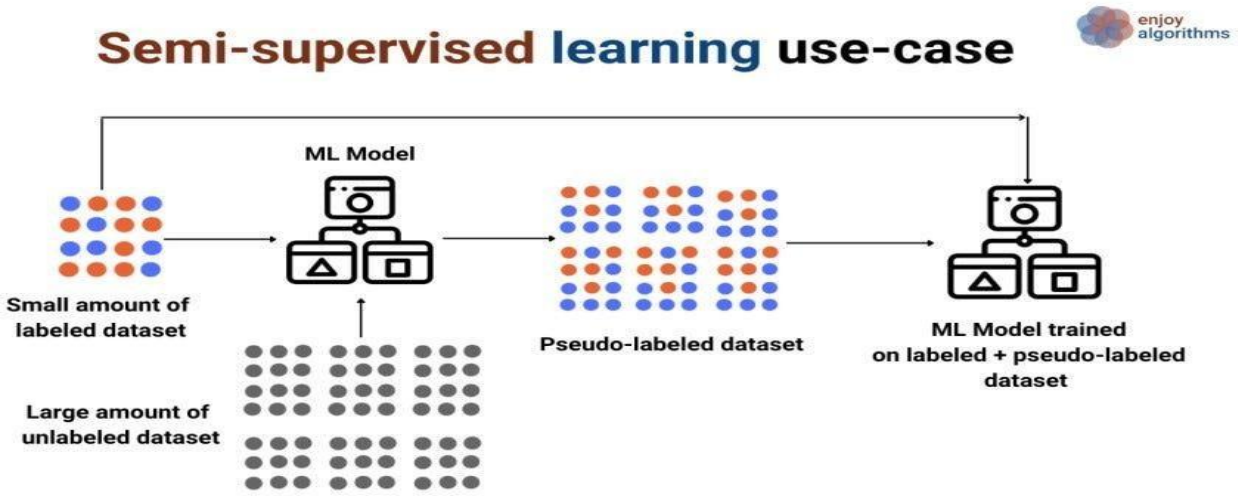


Fig 1.16: Semisupervised Learning-Use Case

For example, suppose there is a large chunk of data in the image above, and a small amount of labelled dataset is present. We can train the model using that small amount of labelled data and then predict on the un labelled dataset. Prediction on an unlabeled dataset will attach the label with every data sample with little accuracy, termed a Pseudo-labeled dataset. Now we can train a new model with a mixture of the true-labelled and pseudo-labelled datasets.

Unsupervised Learning: We can capture and learn the structure present in the data.

Reinforcement Learning:

Here, Machine Learning algorithms act as virtual agents in the known environment where these agents choose the possible action options. The agent selects the best action from all the possibilities present in that environmental state and, based on that selection, receives reward/risks. The algorithms keep an eye on maximizing the reward, reducing the risk, and eventually learning.

Let's take one example. Suppose we want our computer to learn "How to balance the pole on the moving cart"? We might have played this game in childhood, balancing sticks on our palms.



Fig 1.17: Semisupervised Learning-Sample

Here, the hand is replaced by a cart. Suppose we want to make our cart smart enough to balance the stick. So "cart" is the agent, the plane on which the cart will move is the environment, and the cart is taking possible actions such as moving either left or right to balance the stick. So, whenever a stick falls in either direction, the cart will act appropriately to make it stand. Suppose we told our agent that the longer it holds the stick upright, the higher the reward will be.

After every action, the state of the cart and the stick in the environment will change. Our agent will analyze that state and again take different actions best suitable for that particular state.



Fig 1.18: Semisupervised Learning-Illustration

Some famous use cases of Reinforcement Learning are:

If we look at the history of machine learning, we will find that RL is quite old and hasbeen in the industry for a more extended period. But because of the requirement ofawareness of the entire environment states, it is usually used with simulated environments.

Some of the most common use cases where it is being used in the industry are:

- The agent can drive the vehicle inside the simulated environment.
- Predicting the stock price in the stock market.
- Agents that can play games.

Some Frequently used algorithms in Reinforcement Learning :

- Q-Learning
- Deep Q-Learning (DQN)

Table 1.5: Comparison of Machine Learning Model

Type	Supervised	Unsupervised	Semi-supervised
Input Data	Labelled	Unlabelled	Partially Labelled
Computational Complexity	Simpler	Computationally Complex	Depends on use of Supervised or unsupervised
Accuracy	Higher	Lesser	Lesser

6.Data Science as a Discipline (2010s-Present)

- **Interdisciplinary Field:** Data science emerged as a distinct field, combining aspects of statistics, computer science, and domain-specific knowledge. Universities started offering dedicated data science programs.
- **Data-Driven Decision Making:** Companies began to embed data science into their decision-making processes, leveraging insights for competitive advantage.
- **Example:** Netflix's use of data science for content recommendation, personalization, and optimizing content production based on viewer data.

7. Current Trends and Future Directions

- **Deep Learning and AI:** The focus is shifting towards deep learning and AI, with applications in natural language processing (NLP), computer vision, and autonomous systems.
- **Edge Computing:** With IoT devices generating vast amounts of data, edge computing is becoming important for real-time data processing closer to the data source.
- **Ethics and Privacy:** As data science impacts more aspects of life, issues related to data privacy, security, and ethical AI are becoming more prominent.
- **Example:** Autonomous vehicles like Tesla using deep learning for real-time decision making and navigation.

❖ Data Science Project Stages:

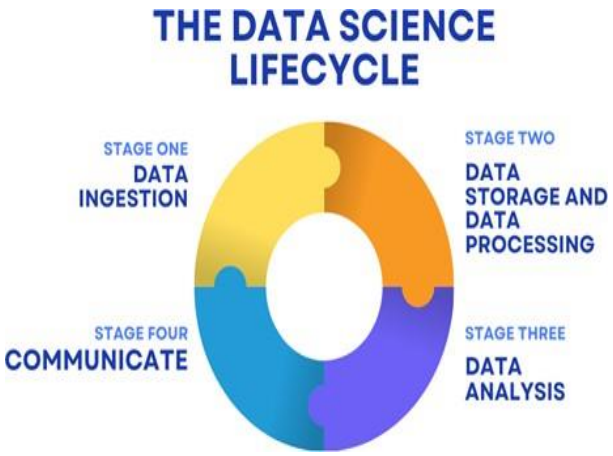


Fig 1.19: Data Science Lifecycle

- **Data ingestion:** Data collection—both raw structured and unstructured data from all relevant sources using a variety of methods.
- **Data storage and data processing:** Data management teams help to set standards around data storage and structure, which facilitate workflows around analytics, machine learning and deep learning models. This stage includes cleaning data, deduplicating, transforming and combining the data using ETL (extract, transform, load) jobs or other data integration technologies.
- **Data analysis**: Data scientists conduct an exploratory data analysis to examine biases, patterns, ranges, and distributions of values within the data. This data analytics exploration drives

hypothesis generation for a/b testing. It also allows analysts to determine the data's relevance for use within modeling efforts for predictive analytics, machine learning, and/or deep learning.

- **Communicate:** Finally, insights are presented as reports and other data visualizations that make the insights—and their impact on business—easier for business analysts and other decision-makers to understand. A data science programming language such as **R or Python** includes components for generating visualizations; alternately, data scientists can use dedicated visualization tools

2. DATA SCIENCE ROLES

1. Data Scientist:

The most obvious key role in a data science team is that of the data scientist. A data scientist is inherently very curious **– trying to understand certain phenomena through the analysis of modeling of complex data.**

Among all the team roles, the data scientist tends to be the **strongest in statistics, math, and machine learning.** They should also have a strong foundation in programming – typically in **Python or R**. Often they start their careers or studies in math or a quantitative research-oriented such as economics or physics. The role generally is not entry-level and might require an advanced degree and a few years of experience.

Typical Responsibilities:

Develop statistical models, machine learning algorithms, and predictive analytics solutions to address business challenges.

Analyze large amounts of complex data to extract insights and drive decision-making.

Design experiments to test hypotheses and measure the effectiveness of solutions.

Collaborate with data engineers and data analysts to collect and preprocess data, and build and maintain data pipelines.

Use data visualization tools to communicate insights and findings to stakeholders.

Skills: Strong background in statistics, programming (Python, R), machine learning, data visualization, and domain knowledge.

Example:

Credit Scoring Model:

In the finance industry, predictive models are used to assess the creditworthiness of individuals applying for loans or credit cards. The model uses historical data such as past credit history, income, employment status, and other factors to predict the likelihood that an individual will default on a loan.

Steps to Create a Predictive Model:

1. **Data Collection:** Gather historical data relevant to the prediction task.
2. **Data Preprocessing:** Clean and preprocess the data, handle missing values, and transform variables if necessary.
3. **Feature Selection:** Select the most relevant features that influence the prediction.
4. **Model Selection:** Choose an appropriate predictive modeling technique (e.g., linear regression, decision trees, neural networks).
5. **Training the Model:** Use the historical data to train the model.

6. **Model Evaluation:** Evaluate the model's performance using metrics like accuracy, precision, recall, and F1-score.
7. **Prediction:** Apply the trained model to new data to make predictions.

Example in Healthcare:

Disease Prediction: A predictive model can be used to predict the likelihood of patients developing certain diseases based on their medical history, lifestyle factors, and genetic information. For instance, a model might predict the risk of diabetes by analyzing data such as age, weight, blood sugar levels, and family history.

By using predictive models, organizations can make data-driven decisions, anticipate future trends, and improve outcomes in various domains.

2. Data Analyst

A data analyst is a professional who is responsible for collecting, processing, and performing statistical analyses on large sets of data. They use various analytical tools and techniques to extract meaningful insights from data and communicate those insights to decision-makers. Often this role overlaps with similar functions such as business analysts who lead requirements gathering and understand business needs and business intelligence analysts who build dashboards for stakeholder consumption.

This role is similar to a data scientist but data analysts tend to be more focused on reporting on the current state as opposed to predictive analytics. Knowing Python and R helps but data analysts are generally not as skilled in programming as data scientists. Many organizations embed data analysts withinbusiness function teams.

Typical Responsibilities:

- Collect and preprocess data from multiple sources to ensure data quality, accuracy, and consistency.
- Analyze and interpret complex data to identify patterns and trends, and to provide insights that support business decision-making.
- Develop dashboards and reports using data visualization tools to communicate insights and findings to stakeholders.
- Collaborate with data scientists and data engineers to collect and preprocess data, and build and maintain data pipelines.

Skills: Proficiency in SQL, Excel, data visualization tools (Tableau, Power BI), and basic statistical knowledge.

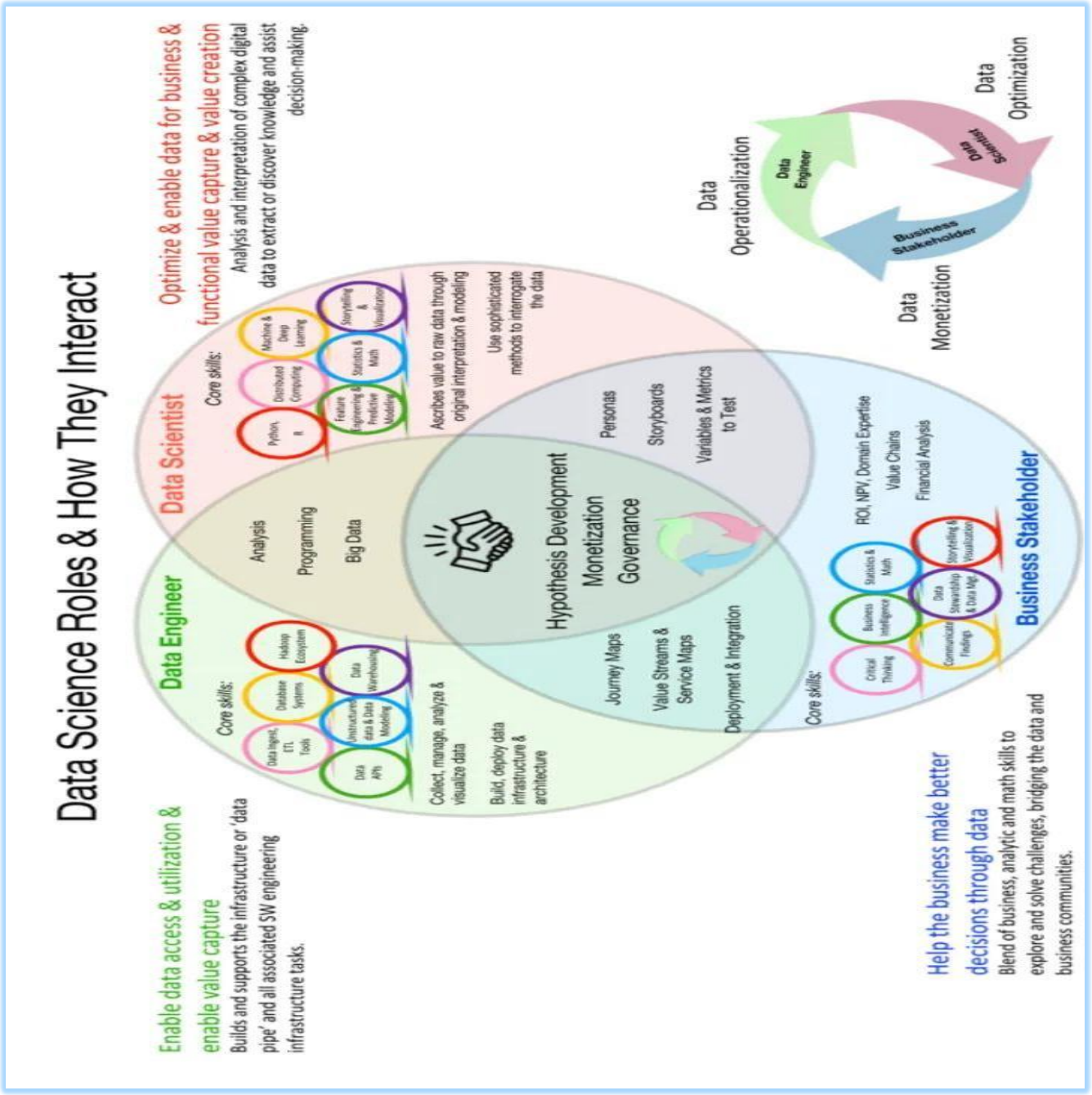


Fig 1.20: Data Science Roles

3. Data Engineer

- Without good data – you can't get very far in data science. Thus, perhaps the second most key role is that of data engineer who is responsible for the collection, storage, and processing of data.
- They design, build, and maintain the infrastructure that enables the data science team to work with large amounts of data. This includes databases, data pipelines, and data warehousing solutions. The data engineer ensures that data is available when and where it is needed, and that it is of high quality. Some organizations have their data engineers sit in a separate team from the data scientists..

Typical Responsibilities:

- Design, build, and maintain data pipelines to move and transform data from various sources into a target location such as a data warehouse or data lake.
- Develop and maintain the infrastructure required to support data science initiatives, including data warehousing, ETL or ELT tools, and data integration solutions.
- Ensure data quality, accuracy, and consistency across multiple data sources.
- Work with data scientists, data analysts, and other stakeholders to understand data requirements and provide support for data-driven decision-making.

Skills: Knowledge of ETL processes, database systems (SQL, NoSQL), big data technologies (Hadoop, Spark), and programming (Python, Java, Scala).

4. Machine Learning Engineer

- This role blends the world of software (and typically cloud) engineering and data science. The machine learning engineer is responsible for building and deploying machine learning models. They work closely with the data scientist to determine the best algorithms and models to use, and they build and implement these models in a production environment. The machine learning engineer is also responsible for monitoring the performance of models and making updates and improvements as necessary.
- Relative to the data scientist, the machine learning engineer tends to be weaker is math/stats but stronger in writing production code and maintaining production systems.
- Sometimes, this role sits outside of the data science team on a software or data engineering team.

Typical Responsibilities:

- Design, develop, and deploy scalable machine learning models and systems that support business objectives.
- Collaborate with data scientists and data engineers to collect and preprocess data, and build and maintain data pipelines.
- Develop and maintain data infrastructures that support machine learning workflows, including data storage, feature engineering, and model training.
- Design and implement distributed systems that support large-scale machine learning.
- Develop and maintain machine learning workflows that are efficient, reproducible, and scalable.
- Implement monitoring and evaluation systems that track model performance and identify potential issues.

Skills: Strong programming skills (Python, Java), understanding of machine learning frameworks (TensorFlow, PyTorch), and knowledge of software engineering practices.

5. Data Architect

Data architects are IT professionals who are tasked with defining policies, procedures, models, and technologies that will be used to collect, organize, store, and retrieve information for the organization.

A data architect is an expert who formulates the organizational data strategy, including standards of data quality, the flow of data within the organization, and security of data. It's the vision of this data management professional that converts business requirements into technical requirements.

As the critical link between business and technology, the demand for qualified data architects has been on the rise.

Typical Responsibilities:

- Developing and implementing an overall organizational data strategy that is in line with business processes. The strategy includes data model designs, database development standards, implementation and management of data warehouses and data analytics systems.
- Identifying data sources, both internal and external, and working out a plan for data management that is aligned with organizational data strategy.
- Coordinating and collaborating with cross-functional teams, stakeholders, and vendors for the smooth functioning of the enterprise data system.
- Managing end-to-end data architecture, from selecting the platform, designing the technical architecture, and developing the application to finally testing and implementing the proposed solution.
- Planning and execution of big data solutions using technologies such as Hadoop. In fact, the big data architect roles and responsibilities entail the complete life-cycle management of a Hadoop Solution.
- Defining and managing the flow of data and dissemination of information within the organization.
- Integrating technical functionality, ensuring data accessibility, accuracy, and security.
- Conducting a continuous audit of data management system performance, refine whenever required, and report immediately any breach or loopholes to the stakeholders.

Skills: Expertise in database design, data modeling, data warehousing, and knowledge of data governance and security practices.

6. Business Intelligence (BI) Developer

- A Business Intelligence Developer is a software engineer specialized in developing, deploying, and maintaining BI interfaces, including data visualization tools, data analytics, and reporting software. Their work involves transforming data into insights that drive business decisions. BI Developers are critical in enhancing business efficiency and productivity by providing the most relevant and current data for decision-making processes.

Typical Responsibilities:

- Create and implement BI software and systems, including integration with databases and data warehousing.
- Develop dashboards, reports, and visualizations to present complex data in an understandable and actionable format.

- Build and maintain data models to ensure the accuracy and relevancy of the data. Manage database performance, security, and integrity.
- Oversee data extraction, transformation, and loading from various sources into a usable form for business analysis.
- Work closely with stakeholders to understand their data needs and translate business requirements into technical solutions.
- Provide ongoing support and training to users on BI tools and data analytics.
- Perform regular maintenance and updates to existing BI solutions to enhance functionality and incorporate new data sources or analytics capabilities.
- Ensure the accuracy, completeness, and reliability of the data used for analysis. Implement data governance policies and practices.
- Stay ahead of the latest trends and technologies in BI, data analytics, and data visualization to improve our BI capabilities continuously.

Skills: Proficiency in BI tools (Tableau, Power BI), SQL, data modeling, and understanding of business processes and requirements.

6. Data Consultant

- As a data consultant, you make sense of large bodies of data so that a company can understand the market and its performance and make important business decisions. In addition to understanding data, you will most likely develop data strategies that align with business objectives to drive companies forward.
- While working as a data consultant, you might coordinate with a group of experts who collect and analyze data. Other tasks in your position could include designing data management and storage systems and communicating with relevant departments. To work in this role, you will most likely need excellent data-analysis skills and experience and the ability to lead and communicate well with others.

Typical Responsibilities:

- Designing and implementing data management systems
- Assessing and improving data analysis processes
- Developing and using data models
- Making recommendations on changes in data analysis and management
- Analyzing business aims and aligning them to data strategies
- Training staff on data analysis and reporting
- Producing data analysis reports
- Using analytics software and systems

- Adhering to data governance procedures
- Liaising with executive-level staff and making recommendations based on data

Skills: Strong analytical and problem-solving skills, knowledge of data technologies, and the ability to communicate effectively with stakeholders.

7. Data Visualization Specialist

- A Data Visualization specialist is the person taking care of such functions as turning plain data into interactive and creative visualizations, after which they are converted to visual elements. Specialists with the ability to close the data processing and decision-making loop by designing interactive visualizations, which are the easy-to-understand information components to be used in a clear strategic manner.
- Showing data through special techniques and methods is considered the main duty in this job; it's more like every data should be shown clearly and attractively. This is primarily about meeting the data needs, data clean up and preparing the data, visualization design and development, and underlying insight presentation.

Typical Responsibilities:

- **Data Analysis and Preparation:**
  - ✓ Understand data and objectives.
  - ✓ Clean, transform, and refine data.
  - ✓ Ensure correct and consistent visualizations.
- **Visualization Design:**
  - ✓ Create graphics, diagrams, maps, charts, and dashboards.
  - ✓ Use tools like Tableau, Power BI, or D3.js.
  - ✓ Apply design methods and color theory.
- **Stakeholder Collaboration:**
  - ✓ Act as a liaison with corporate stakeholders.
  - ✓ Collect requirements and define essential data.
  - ✓ Provide useful insights to decision-makers.
- **Data Interpretation:**
  - ✓ Identify trends, patterns, correlations, and outliers.
  - ✓ Summarize findings through visual storytelling.
  - ✓ Provide practical recommendations to stakeholders.
- **Quality Assurance:**
  - ✓ Ensure data accuracy and thoroughness.
  - ✓ Perform quality checks and data verification.
  - ✓ Fix discrepancies and maintain reliable data sources.

Skills: Proficiency in data visualization tools (Tableau, D3.js), graphic design principles, and an understanding of data storytelling.

8. Chief Data Officer (CDO)

- The CDO is responsible for managing the company's data as a strategic asset, governing its quality and security, and driving business value through data analytics and insights. Moreover, from a cybersecurity and enterprise risk management perspective, having a CDO can greatly enhance an organization's data protection capabilities and risk mitigation strategies.

Typical Responsibilities:

- The CDO oversees an organization's data strategy and ensures that data is used effectively to drive business outcomes. They manage data governance, data quality, and data-related initiatives.

Skills: Leadership and management skills, deep understanding of data strategy, data governance, and the ability to align data initiatives with business goals.

These roles work together to manage and analyze data, develop models and algorithms, and provide insights that help organizations make data-driven decisions. Each role requires a unique combination of skills and expertise, contributing to the overall success of data science projects.

3. VARIOUS FIELDS OF APPLICATION

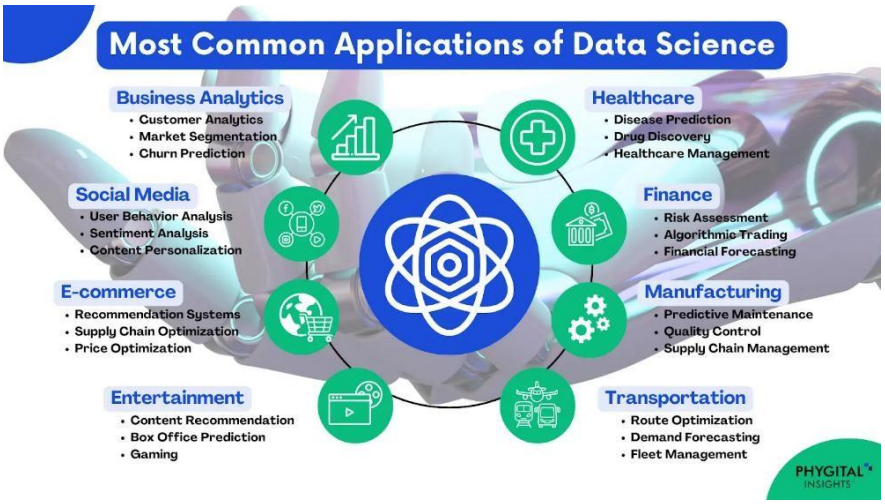


Fig 1.21: Data Science Applications

Data science is a multidisciplinary field that combines statistical techniques, computational algorithms, and domain knowledge to extract insights and knowledge from structured and unstructured data. Here are some of the various fields of application for data science:

1. Healthcare:

- Predictive analytics for patient outcomes.
- Personalized medicine and treatment plans.
- Medical image analysis.

- Genomic data analysis.
- Epidemic outbreak prediction.

2. Finance:

- Fraud detection and prevention.
- Risk management.
- Algorithmic trading.
- Credit scoring.
- Customer segmentation.

3. Retail:

- Customer behavior analysis.
- Recommendation systems.
- Inventory management.
- Price optimization.
- Market basket analysis.

4. Marketing:

- Targeted advertising.
- Customer segmentation and profiling.
- Campaign effectiveness analysis.
- Sentiment analysis.
- Churn prediction.

5. Manufacturing:

- Predictive maintenance.
- Quality control.
- Supply chain optimization.
- Process optimization.
- Demand forecasting.

6. Transportation and Logistics:

- Route optimization.
- Predictive maintenance of vehicles.
- Traffic management.
- Autonomous vehicles.
- Supply chain logistics.

7. Energy:

- Smart grid management.
- Predictive maintenance of equipment.
- Energy consumption forecasting.
- Renewable energy optimization.
- Oil and gas exploration.

8. Education:

- Personalized learning plans.
- Student performance prediction.
- Curriculum development.
- Dropout prediction.
- Learning analytics.

9. Entertainment:

- Content recommendation systems.
- Audience sentiment analysis.
- Box office revenue prediction.
- Social media analytics.
- Game development and player behavior analysis.

10. Telecommunications:

- Network optimization.
- Customer churn analysis.
- Fraud detection.
- Call detail record analysis.
- Predictive maintenance of network infrastructure.

11. Public Sector:

- Crime prediction and prevention.
- Public health monitoring.
- Disaster response and management.
- Urban planning.
- Policy impact analysis.

12. Sports:

- Player performance analysis.
- Injury prediction and prevention.
- Game strategy optimization.
- Fan engagement.
- Sports betting analysis.

13. Agriculture:

- Crop yield prediction.
- Precision farming.
- Soil health monitoring.
- Pest and disease prediction.
- Supply chain optimization.

14. Environmental Science:

- Climate change modeling.
- Wildlife conservation.
- Pollution monitoring.
- Natural disaster prediction.
- Resource management.

15. Human Resources:

- Employee performance analysis.
- Talent acquisition and recruitment.
- Workforce planning.
- Employee sentiment analysis.
- Retention and turnover analysis.

Data science continues to evolve and expand into new areas as data becomes more integral to decision-making processes across various industries.

4. TYPES OF DATA-INTRODUCTION AND TERMINOLOGY-CLASSIFICATION AND CHARACTERISTICS OF DATA TYPES.

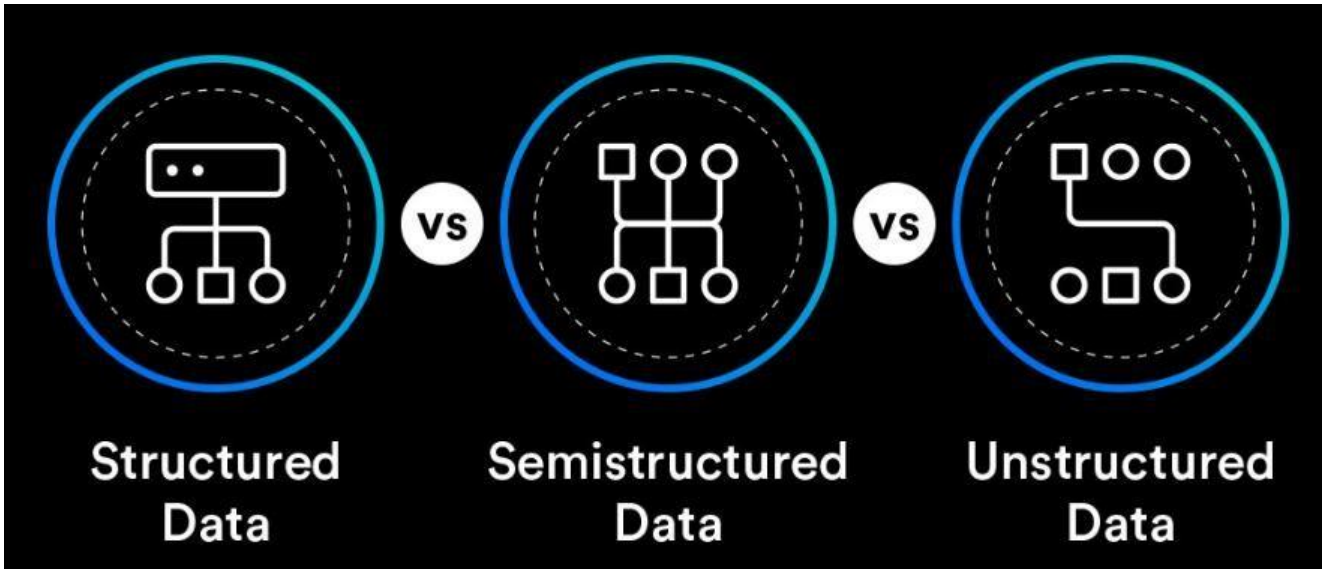


Fig 1.22: Data Structure Classification

5. UNDERSTANDING STRUCTURED DATA

Structured Data is data that is highly organized and easily searchable in fixed fields within a file (e.g., spreadsheets) or database (e.g., SQL databases). This type of data is often formatted in tables with rows and columns, making it straightforward to analyze and process using traditional data management tools and methods.

Characteristics of Structured Data:

- **Tabular Format:** Organized into rows and columns.
- **Defined Data Types:** Each column has a specific data type (e.g., integer, string, date).
- **Easily Searchable:** Can be easily queried and indexed.
- **Schema:** Adheres to a strict schema, which defines the structure and types of data.

Examples of Structured Data:

1. **Relational Databases:**
   - **Example:** SQL databases like MySQL, PostgreSQL, and Oracle.
   - **Description:** Data is stored in tables. Each table has rows (records) and columns (fields). Tables can be related to each other through foreign keys.
   - **Use Case:** Customer databases, inventory management systems.
2. **Spreadsheets:**
   - **Example:** Microsoft Excel, Google Sheets.
   - **Description:** Data is organized into rows and columns within a spreadsheet file.
   - **Use Case:** Budget tracking, sales reports, project management.
3. **CSV Files:**

- **Example:** A file containing data like "Name, Age, Email\nJohn Doe, 30, john.doe@example.com".
- **Description:** Comma-separated values files are plain text files where each line represents a row and each value is separated by a comma.
- **Use Case:** Exporting and importing data between systems, data exchange.

4. **XML Files:**
   - **Example:** A file containing data like `<person><name>John Doe</name><age>30</age><email>john.doe@example.com</email></person>`.
   - **Description:** Extensible Markup Language files use tags to define data structure and relationships.
   - **Use Case:** Web services, configuration files.
5. **JSON Files:**
   - **Example:** A file containing data like `{"name": "John Doe", "age": 30, "email": "john.doe@example.com"}`.
   - **Description:** JavaScript Object Notation files use key-value pairs to represent data structures.
   - **Use Case:** APIs, data interchange formats for web applications.

Example of Structured Data in a Table (Relational Database):

CustomerID	Name	Age	Email
1	John Doe	30	john.doe@example.com
2	Jane Smith	25	jane.smith@example.com
3	Bob Brown	45	bob.brown@example.com

In this table:

- Each row represents a customer.
- Each column represents a specific attribute of the customer (CustomerID, Name, Age, Email).
- The data is organized and easily searchable through queries.

Advantages of Structured Data:

- **Efficiency:** Easy to store, retrieve, and manage using SQL queries.
- **Clarity:** Clear schema makes data interpretation straightforward.
- **Data Integrity:** Enforced through constraints and relationships in databases.
- **Compatibility:** Well-supported by various data analysis and business intelligence tools.

Applications of Structured Data:

- **Business Intelligence:** Generating reports, dashboards, and insights.
- **Data Warehousing:** Centralized storage for analysis and reporting.
- **CRM Systems:** Managing customer information and interactions.
- **Financial Systems:** Tracking transactions, accounts, and audits.

Structured data is fundamental in many industries for effective data management and analysis due to its organized and easily accessible nature.

6. UNDERSTANDING UNSTRUCTURED DATA

Unstructured Data is data that does not have a predefined format or organization. It is not easily searchable using traditional data management tools and often requires more sophisticated techniques for storage, processing, and analysis. This type of data includes a wide variety of content such as text, images, audio, and video.

Characteristics of Unstructured Data:

- **No Fixed Schema:** Lacks a specific structure or format.
- **Varied Types:** Can include text, multimedia, social media posts, emails, etc.
- **Complexity:** Often requires advanced processing techniques like natural language processing (NLP) or machine learning to analyze.
- **Large Volume:** Typically involves large volumes of data.

Examples of Unstructured Data:

1. **Text Documents:**
   - **Example:** Word documents, PDFs.
   - **Description:** Written content without a predefined structure.
   - **Use Case:** Contracts, reports, research papers.
2. **Emails:**
   - **Example:** Emails with varying formats including attachments.
   - **Description:** Communication content with text, metadata (sender, recipient, timestamp), and attachments.
   - **Use Case:** Customer support interactions, business communications.
3. **Social Media Posts:**
   - **Example:** Tweets, Facebook posts, Instagram captions.
   - **Description:** User-generated content with text, hashtags, mentions, and sometimes multimedia elements.
   - **Use Case:** Sentiment analysis, brand monitoring, social media marketing.
4. **Multimedia Files:**
   - **Example:** Images (JPEG, PNG), videos (MP4, AVI), audio files (MP3, WAV).
   - **Description:** Visual and auditory content without a fixed structure.
   - **Use Case:** Media archives, content creation, video surveillance.
5. **Web Pages:**
   - **Example:** HTML pages, blogs, news articles.
   - **Description:** Web content with a mix of text, images, and multimedia, often with varied formatting.
   - **Use Case:** Web scraping, SEO analysis, content aggregation.
6. **Sensor Data:**
   - **Example:** Data from IoT devices like temperature sensors, motion detectors.
   - **Description:** Continuous streams of data in various formats.
   - **Use Case:** Smart home systems, industrial monitoring, environmental sensing.

Example of Unstructured Data:

Email Example:

From: john.doe@example.com
To: jane.smith@example.com
Subject: Project Update
Hi Jane,
I wanted to give you an update on the project. We have completed the initial phase and are now moving on to the next steps. Please find the attached report for detailed information.
Best, John

In this email:

- The content includes text with a natural language structure.
- Metadata includes the sender, recipient, and subject.
- Attachments may include documents or images.
- The structure and format vary greatly between different emails.

Advantages of Unstructured Data:

- **Richness:** Contains a wealth of information that can provide deeper insights.
- **Flexibility:** Can capture diverse types of data without requiring a rigid schema.
- **Real-World Context:** Often reflects real-world scenarios and user behavior more accurately.

Applications of Unstructured Data:

- **Customer Feedback Analysis:** Extracting insights from reviews, surveys, and social media comments.
- **Content Management:** Organizing and retrieving multimedia content in digital libraries.
- **Natural Language Processing:** Understanding and processing human language in text documents.
- **Image and Video Analysis:** Detecting objects, faces, and activities in multimedia files.
- **Big Data Analytics:** Analyzing large volumes of varied data for trends and patterns.

Unstructured data is increasingly important in today's data-driven world, offering valuable insights that structured data alone cannot provide. Advanced technologies and techniques are continually being developed to harness the potential of unstructured data effectively.

7. UNDERSTANDING SEMI STRUCTURED DATA

Semi-Structured Data is a form of data that does not conform to the rigid structure of traditional databases but still contains tags or markers to separate semantic elements and enforce hierarchies of records and fields. It falls between structured and unstructured data, providing some degree of organization that makes it easier to analyze than completely unstructured data.

Characteristics of Semi-Structured Data:

- **Flexible Structure:** Does not adhere to a fixed schema but has a self-describing structure (e.g., tags, attributes).
- **Hierarchical:** Often organized in a hierarchical or nested format.
- **Readable:** Easier to parse and interpret than unstructured data but more flexible than structured data.
- **Interoperability:** Can be easily shared across different systems.

Examples of Semi-Structured Data:

1. **XML (eXtensible Markup Language):**
    - **Example**

```
<person>
  <name>John Doe</name>
  <age>30</age>
  <email>john.doe@example.com</email>
</person>
```

- • **Description:** Uses tags to define elements and their relationships, allowing for a flexible, hierarchical structure.
- **Use Case:** Web services, configuration files, data interchange formats.

- **JSON (JavaScript Object Notation):**

    - **Example**

```
{
  "name": "John Doe",
  "age": 30,
  "email": "john.doe@example.com"
}
```

- **Description:** Uses key-value pairs to represent data, commonly used in web APIs and configuration files.

- **Use Case:** Data interchange between web applications, APIs, NoSQL databases.

- **HTML (HyperText Markup Language):**
  - **Example:**

```
<html>
  <head>
    <title>Sample Page</title>
  </head>
  <body>
    <h1>Welcome to my website</h1>
    <p>My name is John Doe.</p>
  </body>
</html>
```

  - Description**:** Uses tags to define the structure and content of web pages.
  - **Use Case:** Web development, content management systems.

- YAML **(YAML Ain't Markup Language):**
  - **Example**

```
person:
 name: John Doe
 age: 30
 email: john.doe@example.com
```

  - **Description:** A human-readable data serialization standard that is commonly used for configuration files.
  - **Use Case:** Configuration files for software applications, data exchange.

- **Email Headers:**
  - **Example:**

```
From: john.doe@example.com
To: jane.smith@example.com
Subject: Meeting Reminder
Date: Mon, 5 Jul 2021 14:30:00 +0000
```

  - **Description:** Contains structured metadata (headers) and unstructured content (body).
  - **Use Case:** Email systems, communication records.

Example of Semi-Structured Data in JSON Format:

```
{
   "students": [
      {
         "id": 1,
         "name": "Alice",
```

```
            "age": 23,
            "courses": ["Math", "Physics"]
        },
        {
            "id": 2,
            "name": "Bob",
            "age": 22,
            "courses": ["History", "Literature"]
        }
    ]
}
```

In this JSON example:

- **Data is organized** into key-value pairs.
- **Nested structure** represents the relationship between different elements (students and their courses).

Advantages of Semi-Structured Data:

- **Flexibility:** Allows for a flexible data model that can evolve over time.
- **Interoperability:** Can be easily exchanged and processed across different systems.
- **Readability:** More readable and understandable than purely unstructured data.
- **Compatibility:** Can be converted to structured data when needed, enabling further analysis.

Applications of Semi-Structured Data:

- **Web Development:** HTML and JSON are widely used for building and exchanging web content.
- **APIs:** JSON and XML are standard formats for data interchange in web APIs.
- **Configuration Management:** YAML and JSON are commonly used for configuration files in software development.
- **Data Integration:** Facilitates the integration of data from diverse sources.

Semi-structured data provides a balance between the rigid structure of traditional databases and the flexibility of unstructured data, making it highly useful for modern data storage, exchange, and processing needs.

8. DATA SECURITY ISSUES IN DATA SCIENCE

Data security is a critical concern in data science due to the sensitive nature of the data being processed and analyzed. Ensuring the confidentiality, integrity, and availability of data throughout its lifecycle is paramount to protecting against data breaches, unauthorized access, and other security threats. Here is a detailed explanation of data security issues in data science:

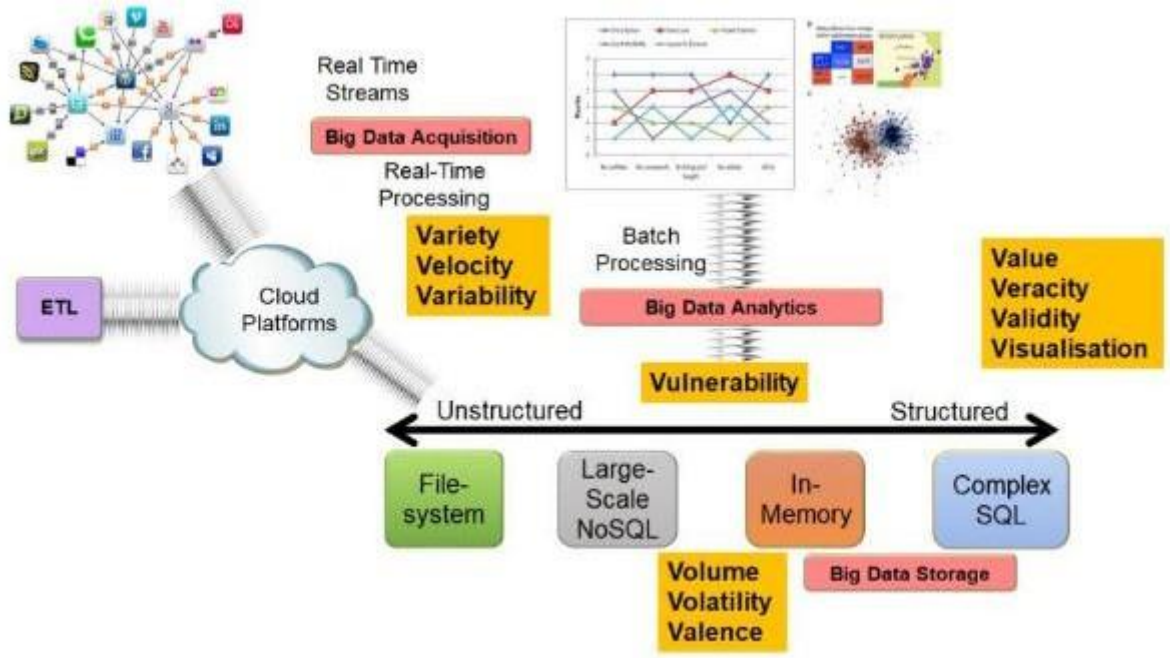


Fig 1.23: Data Science Security Issues

1.Data Breaches

- **Description:** Unauthorized access to sensitive data by malicious actors.
- **Impact:** Can lead to exposure of confidential information, financial losses, and damage to an organization's reputation.
- **Examples:** Hackers stealing customer data, internal employees leaking sensitive information.

2. Data Privacy

- **Description:** Ensuring that personal and sensitive data is collected, processed, and stored in compliance with privacy regulations.
- **Impact:** Violations can result in legal penalties, loss of customer trust, and reputational damage.
- **Examples:** Compliance with GDPR- General Data Protection Regulation , HIPAA – Health Insurance Portability and Accountability, CCPA- California Consumer Privacy Act.

3. Insider Threats

- **Description:** Security risks posed by employees, contractors, or other insiders who have access to sensitive data.
- **Impact:** Insiders can misuse their access to steal or leak sensitive data, intentionally or unintentionally.
- **Examples:** An employee copying sensitive customer information to a personal device.

4. Data Encryption

- **Description:** Protecting data by converting it into a coded format that can only be read by authorized individuals.

- **Impact:** Ensures data remains secure during transmission and storage, protecting it from unauthorized access.
- **Examples:** Encrypting data in transit using SSL-Secure Socket Layer/TLS- Transport Layer Security, encrypting data at rest using AES.

5. Access Controls

- **Description:** Implementing measures to restrict access to data based on user roles and permissions.
- **Impact:** Prevents unauthorized access to sensitive data, ensuring that only authorized personnel can access specific data.
- **Examples:** Role-based access control (RBAC), multi-factor authentication (MFA).

6. Data Masking

- **Description:** Obscuring specific data within a database to protect it from unauthorized access.
- **Impact:** Allows sensitive data to be used for testing or analysis without exposing the actual data.
- **Examples:** Masking credit card numbers, social security numbers.

7. Anonymization and De-Identification

- **Description:** Removing or altering personal identifiers from data sets so that individuals cannot be readily identified.
- **Impact:** Reduces the risk of re-identification of individuals in case of a data breach.
- **Examples:** Removing names and addresses from health records before analysis.

8. Data Integrity

- **Description:** Ensuring the accuracy and consistency of data over its lifecycle.
- **Impact:** Prevents unauthorized modifications and ensures that data remains trustworthy and reliable.
- **Examples:** Using checksums and hashing to verify data integrity, implementing version control.

9. Secure Data Storage

- **Description:** Safeguarding data in storage systems to prevent unauthorized access or data loss.
- **Impact:** Protects data from physical and cyber threats, ensuring its availability and integrity.
- **Examples:** Using secure cloud storage solutions, implementing redundancy and backup systems.

10. Data Governance

- **Description:** Establishing policies and procedures for managing data security, privacy, and integrity.
- **Impact:** Ensures that data management practices align with regulatory requirements and organizational standards.
- **Examples:** Data governance frameworks, data stewardship roles.

Strategies to Mitigate Data Security Issues

1. **Implement Strong Access Controls:**
   - Use role-based access control (RBAC) to restrict access to sensitive data.
   - Enforce multi-factor authentication (MFA) to add an extra layer of security.
2. **Encrypt Data:**
   - Encrypt data at rest and in transit to protect it from unauthorized access.
   - Use strong encryption algorithms such as AES and RSA.
3. **Regularly Update and Patch Systems:**
   - Keep software and systems up to date with the latest security patches.
   - Regularly update security protocols and encryption standards.
4. **Conduct Regular Security Audits:**
   - Perform periodic security audits to identify vulnerabilities and address them promptly.
   - Use penetration testing to simulate attacks and improve defenses.
5. **Data Masking and Anonymization:**
   - Apply data masking techniques to protect sensitive information in non-production environments.
   - Use anonymization methods to remove personally identifiable information from data sets.
6. **Employee Training and Awareness:**
   - Train employees on data security best practices and the importance of data protection.
   - Conduct regular security awareness programs to keep employees informed about the latest threats.
7. **Implement Data Loss Prevention (DLP) Solutions:**
   - Use DLP tools to monitor and protect sensitive data from unauthorized access or transfer.
   - Set up alerts and automated responses to potential data breaches.
8. **Establish a Data Governance Framework:**
   - Develop and enforce data governance policies and procedures.
   - Assign data stewardship roles to ensure accountability and oversight.
9. **Regular Backups and Disaster Recovery Planning:**
   - Implement regular data backups and test recovery procedures.
   - Develop a comprehensive disaster recovery plan to ensure business continuity.
10. **Monitor and Respond to Security Incidents:**
    - Set up continuous monitoring systems to detect and respond to security incidents in real-time.
    - Establish an incident response team to handle data breaches and other security threats.

Data security is a vital aspect of data science, requiring a multi-faceted approach to protect sensitive information from various threats. By implementing robust security measures, adhering to privacy regulations, and fostering a culture of security awareness, organizations can mitigate the risks associated with data security issues and ensure the safe handling of data throughout its lifecycle.

9. EXAMPLE APPLICATIONS SHOWCASING REAL-WORLD SCENARIOS

Real-World Applications of Data Science

Data science is transforming various industries by enabling data-driven decision-making and insights extraction. Here are some detailed examples of real-world applications showcasing how data science is utilized across different sectors:

1. Healthcare: Predictive Analytics and Personalized Medicine

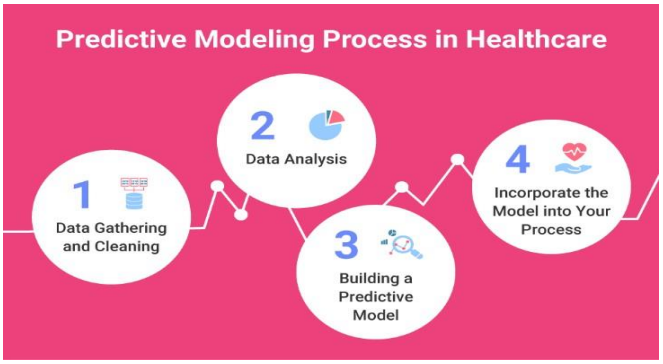


Fig 1.24: Predictive Modeling in Healthcare

- **Description:** Data science techniques such as machine learning and statistical modeling are used to predict patient outcomes and develop personalized treatment plans.
- **Scenario:** Hospitals collect patient data including medical history, genetic information, and lifestyle factors. Machine learning models analyze this data to predict disease risk and recommend personalized treatment plans.
- **Example:** IBM Watson for Oncology uses data science to analyze patient data and medical literature to provide evidence-based treatment options for cancer patients.

2. Finance: Fraud Detection and Risk Management

- **Description:** Financial institutions use data science to detect fraudulent activities and manage financial risks.
- **Scenario:** Banks monitor transaction data in real-time to identify unusual patterns that may indicate fraud. Machine learning models flag suspicious transactions for further investigation.
- **Example:** PayPal uses machine learning algorithms to analyze transaction data and detect fraudulent activities, reducing the incidence of fraud on its platform.



Fig 1.25: Predictive Modeling in Healthcare

3. Retail: Customer Segmentation and Recommendation Systems

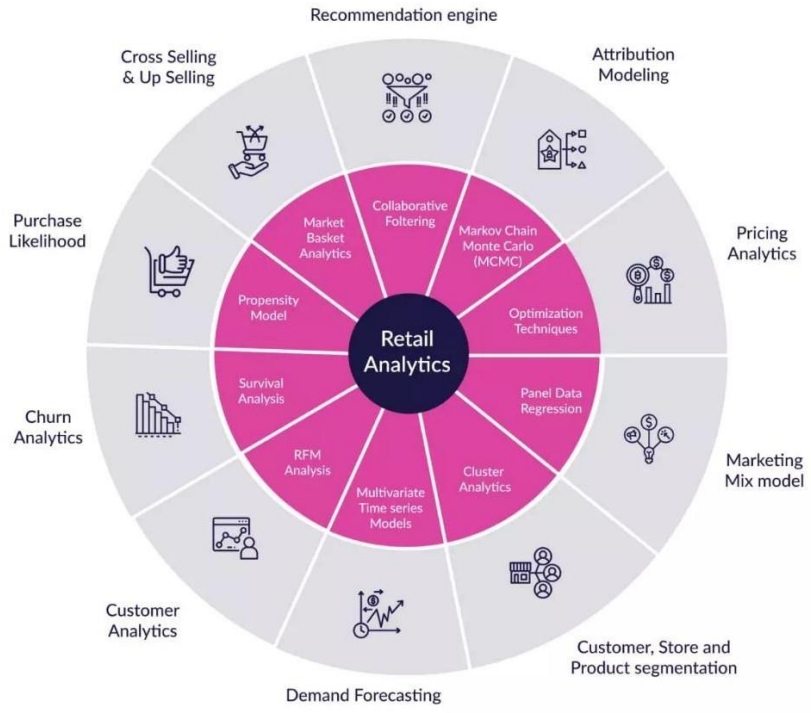


Fig 1.26: Data Science in Retail industry

- **Description:** Retailers leverage data science to segment customers based on purchasing behavior and to provide personalized product recommendations.
- **Scenario:** An e-commerce platform collects data on customer browsing and purchase history. Data science models analyze this data to segment customers and recommend products that match their preferences.

- **Example:** Amazon uses collaborative filtering algorithms to suggest products to customers based on their past purchases and browsing history.

4. Marketing: Sentiment Analysis and Campaign Optimization



Fig 1.27: Data Science in Marketing

- **Description:** Marketers use data science to analyze customer sentiment and optimize marketing campaigns.
- **Scenario:** A company collects social media posts, reviews, and survey responses. Natural language processing (NLP) techniques analyze this text data to gauge customer sentiment and identify trends.
- **Example:** Coca-Cola uses sentiment analysis to monitor customer reactions to its products and marketing campaigns, allowing it to make data-driven decisions on marketing strategies.

5. Manufacturing: Predictive Maintenance and Quality Control

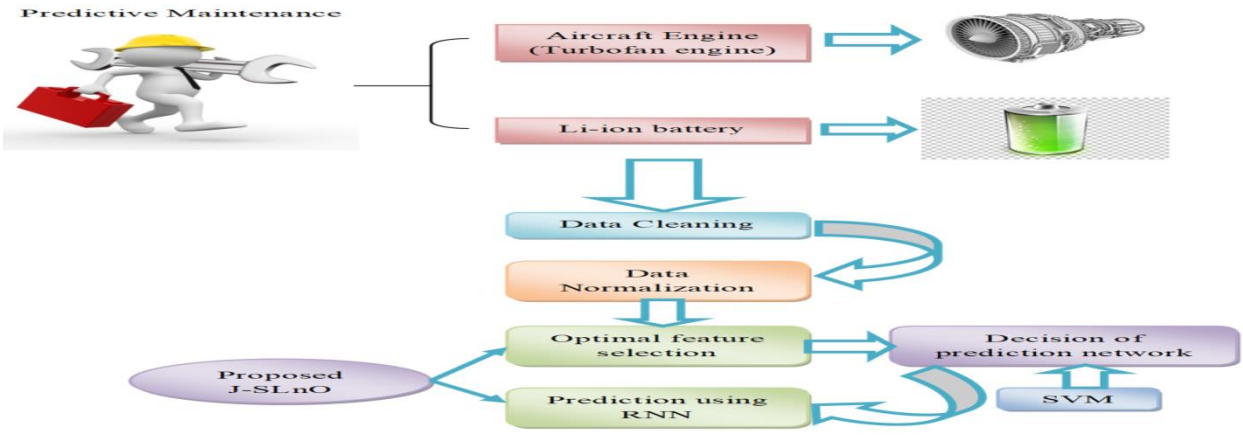


Fig 1.28: Data Science in Manufacturing

- **Description:** Data science helps manufacturers predict equipment failures and ensure product quality.
- **Scenario:** Factories equip machinery with sensors that collect performance data. Machine learning models analyze this data to predict when a machine is likely to fail, allowing for timely maintenance.

- **Example:** General Electric (GE) uses predictive analytics to monitor the health of industrial equipment, reducing downtime and maintenance costs.

6. Transportation: Route Optimization and Traffic Management

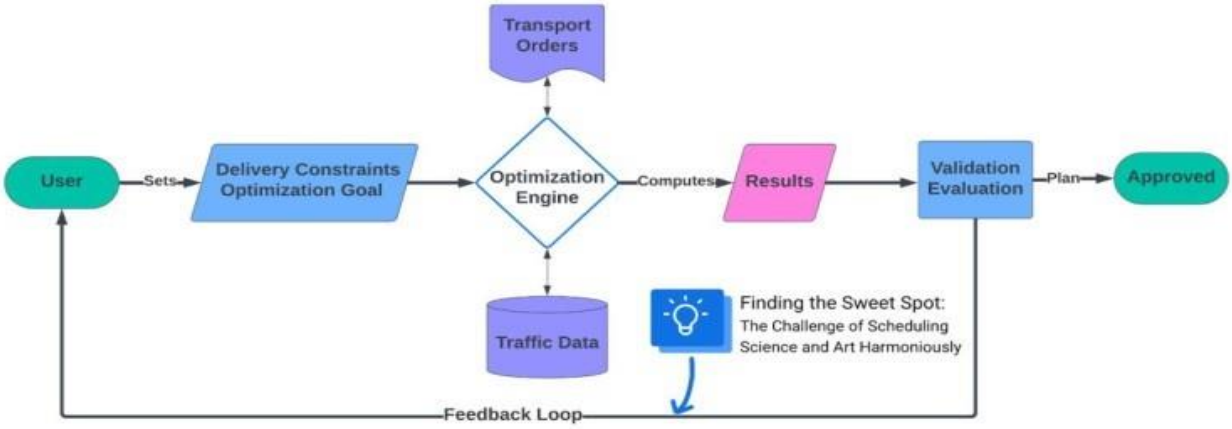


Fig 1.29: Data Science in Transportation

- **Description:** Transportation companies use data science to optimize routes and manage traffic flow.
- **Scenario:** A logistics company collects data on delivery routes, traffic conditions, and vehicle performance. Optimization algorithms calculate the most efficient routes for deliveries.
- **Example:** UPS uses ORION (On-Road Integrated Optimization and Navigation) to optimize delivery routes, saving millions of miles driven and reducing fuel consumption.

7. Energy: Smart Grid Management and Consumption Forecasting

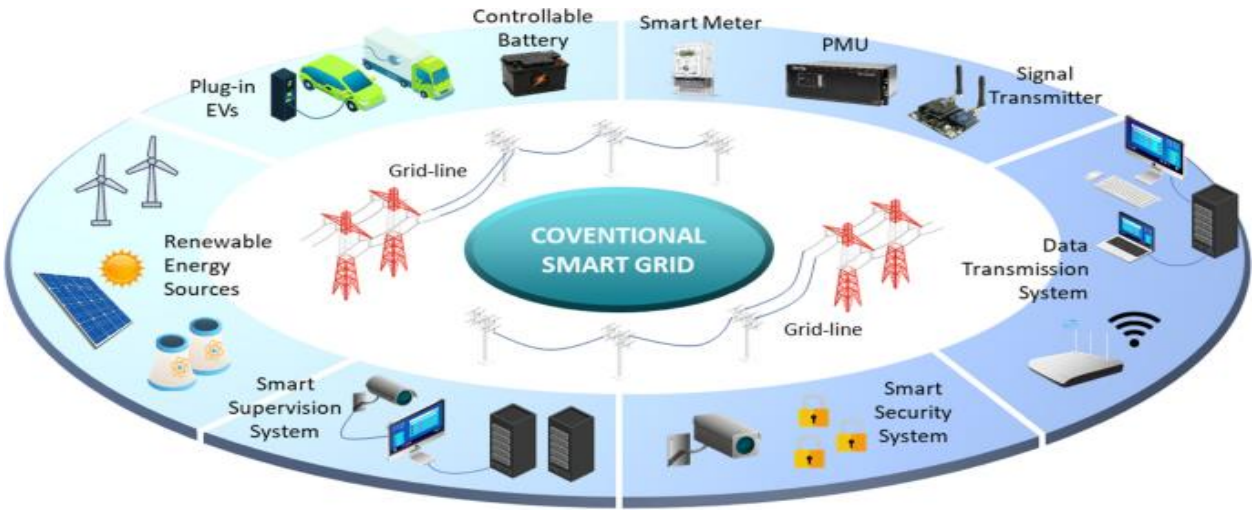


Fig 1.30: Data Science in Energy

- **Description:** Energy companies leverage data science for efficient grid management and to forecast energy consumption.
- **Scenario:** Smart meters collect data on energy usage in real-time. Data science models predict energy demand and help balance the grid by managing supply and demand efficiently.
- **Example:** Pacific Gas and Electric (PG&E) uses machine learning to forecast energy demand, optimizing the distribution of electricity and reducing operational costs.

8. Education: Personalized Learning and Student Performance Analysis

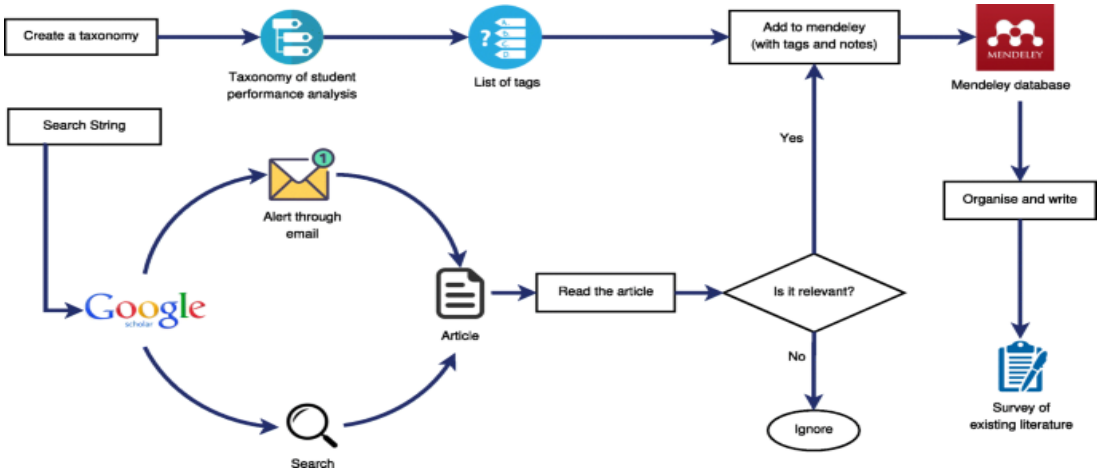


Fig 1.31: Data Science in Education

- **Description:** Educational institutions use data science to tailor learning experiences and analyze student performance.
- **Scenario:** An online learning platform collects data on student interactions, quiz scores, and study habits. Machine learning models personalize the learning content and identify students who need additional support.
- **Example:** Coursera uses data analytics to provide personalized course recommendations and improve student engagement by analyzing their learning patterns.

9. Sports: Performance Analysis and Injury Prediction

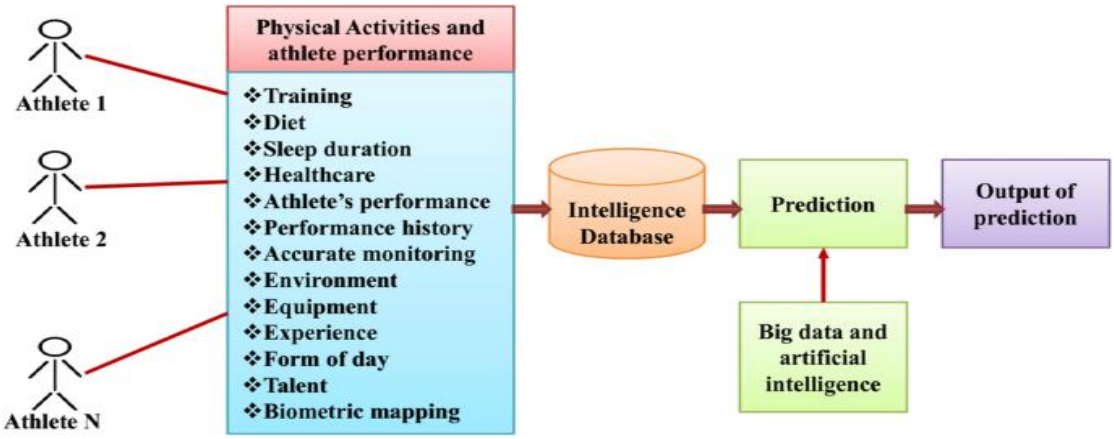


Fig 1.32: Data Science in sports

- **Description:** Sports teams use data science to analyze player performance and predict injuries.
- **Scenario:** Teams collect data on player movements, game statistics, and training sessions. Data science models analyze this data to optimize player performance and reduce the risk of injuries.
- **Example:** Major League Baseball (MLB) teams use Statcast, a data analytics system, to analyze player performance and make strategic decisions during games.

10. Telecommunications: Network Optimization and Customer Churn Prediction

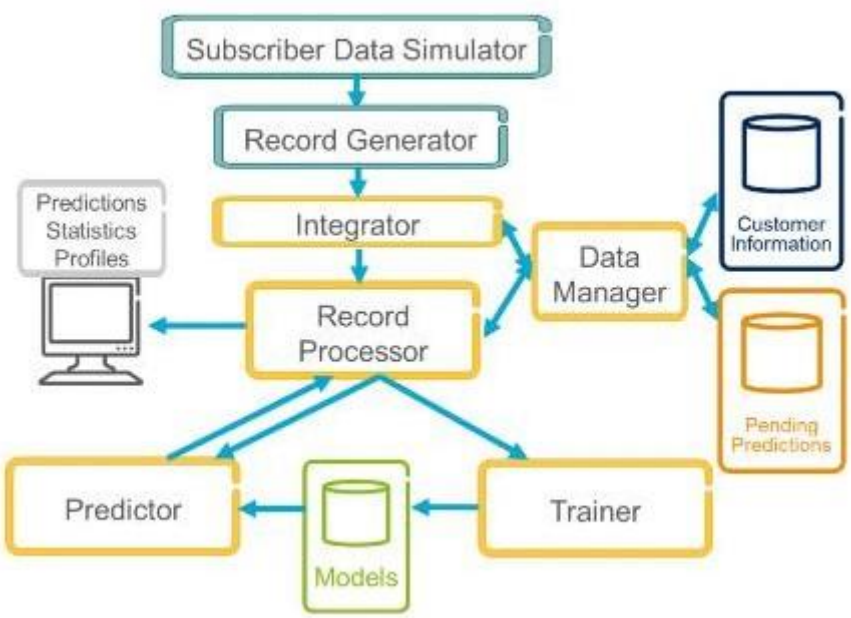


Fig 1.33: Customer Churn Prediction

- **Description:** Telecom companies use data science to optimize network performance and predict customer churn.
- **Scenario:** A telecom company collects data on network usage, customer service interactions, and billing information. Machine learning models predict which customers are likely to switch providers and identify network issues.
- **Example:** Vodafone uses predictive analytics to identify customers at risk of churning and offers targeted promotions to retain them.

Data science is being applied across various industries to solve complex problems, improve efficiency, and provide personalized experiences. By leveraging advanced analytical techniques and machine learning models, organizations can transform raw data into actionable insights, driving innovation and enhancing decision-making processes.